

# A Machine-Checked Finite Reduction for Erdős Problem #287 and a Source-Explicit Large- $M$ Criterion

Clean Public Review + Self-Contained Gate Theory Manual  
Version 16.7 – Public Review R8 (31 August 2026 synchronization)

Research draft with Lean/Aristotle-assisted formal verification

31 August 2026

**Status – 31 August 2026 synchronization. Erdős Problem #287 remains open.**  
The unconditional public Lean theorem excluding exact counterexamples with maximum denominator at most  $4 \times 10^9$  is unchanged. The Gate 0–2 theory manual below is retained as theory rather than replaced by a status-only note.

The newest audited C0 milestone supersedes the controlling 30-August R7 headline. Exact product collision and Double Type II are analytically closed. C0 is analytically closed *conditional on* the separate physical/formal normalisation pin

`SHAREDGCD-OMEGAH-NORMALISATION-SOURCEPIN45`,

together with the complete physical Perron/nuclear normalisation unless absorbed into the same pin. The repaired finite-Fourier contraction is

$$\eta_b^2 \ll L^C \left( \frac{x}{B} + \frac{1}{N} + \frac{1}{E_b} + \frac{1}{x} \right).$$

The product energy is source-specific to the literal bounded dyadic packet, the complete unit branch is  $\gcd(ba_\rho b_\rho u_\rho, x) = 1$ , and small- $x$  packets have an explicit short-conductor/low- $Q$  owner.

The transverse long-conductor branch is strictly reduced but remains **OPEN**. One-conductor reciprocity and the  $q_C$  unitary-Fourier theorem pass at the audited research/paper level; the dense- $q_C$  compiler closes only its stated strict subregion. The naive full-CRT two-signless-carrier DFT is retracted. The current first analytic residual is

`THREEFACTOR-TRANSVERSE-CRITICAL-BEZOUTNUMERATOR-SINGLECARRIER45`.

The parallel  $b$ -diagonal node `BDIAGONAL-AFFINEA-SURVIVING-VERTEX-RECT45` remains separately open.

The supplied 31-August Aristotle bundle records a completed C0 unitary-Fourier / transverse-carrier-interface bank with `lake build`: `PASS`, 8298 jobs and zero errors; its audited declarations use only ordinary Mathlib foundations. The newest named Lean files depend on a larger dependency graph than is presently synchronized in public main, so this manuscript does not pretend that the same 8298-job build has already been reproduced from public `RequestProject/Main.lean`. No unfinished one-conductor formalisation is fabricated or pre-promoted.

## Abstract

Erdős Problem #287 asks whether every representation

$$1 = \frac{1}{n_1} + \cdots + \frac{1}{n_k}, \quad 1 < n_1 < \cdots < n_k,$$

contains a consecutive gap of size at least 3. The strongest unconditional theorem in this project remains machine-checked: there is no exact counterexample with maximum denominator at most  $4 \times 10^9$ . The remaining global range is conditional on explicit source/analytic compilers; no eventual **WindowPairSupply** theorem and no unconditional solution are claimed.

This revision preserves the self-contained Gate 0–2 theory manual and supersedes the controlling frontier/status layer. The 31-August hostile audit closes the canonical C0 exact-product and Double Type II analytic cores, conditional on the separately exposed physical normalisation source pin. After reduced-projective conditioning the relevant Kloosterman kernel is independent of the balanced  $b$  source. Writing  $b = \ell_b d_b e_b$  with  $d_b, e_b > Y$  and grouping  $n = \ell_b d_b$ , residue aggregation and inversion on units produce a restricted finite Fourier matrix with exact norm  $\sqrt{x}$ . Together with fixed-depth source product energy this yields

$$\eta_b^2 \ll L^C \left( \frac{x}{B} + \frac{1}{N} + \frac{1}{E_b} + \frac{1}{x} \right).$$

The statement is source-specific; the complete unit firewall is  $\gcd(ba_\rho b_\rho u_\rho, x) = 1$ , complementary nonunit cells are separately routed, and small- $x$  packets are assigned to the short-conductor/low- $Q$  owner. No conjectural Möbius cancellation and no new spectral black box are used.

For the transverse long-conductor family, the naive two-signless full-CRT DFT is retracted: CRT fusion leaves cross-modulus inverse coefficients in the numerator. A repaired one-conductor reciprocity normal form is banked at the research level. For

$$\Phi_P = e_q(A\overline{r}\overline{m})e_r(B\overline{q}\overline{m}),$$

reciprocating only the  $q$  component gives

$$\Phi_P = \text{Arch}_P e_{m_P}(\Gamma_P^{\text{red}} \overline{q}), \quad \Gamma_P = -A + mB\overline{m} \pmod{r}, \quad m_P = rm/(B - A, r),$$

with  $(\Gamma_P^{\text{red}}, m_P) = 1$ . Grouping  $q = S_1 S_2$  then gives the audited finite-Fourier contraction

$$\eta_P^2 \ll L^C \left( \frac{m_P}{L_1 L_2} + \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{m_P} \right).$$

This closes the dense- $q_C$  subregion under explicit dominance/two-long-group hypotheses, not the full transverse branch.

The first current analytic residual is

$$\boxed{\text{THREEFACTOR-TRANSVERSE-CRITICAL-BEZOUTNUMERATOR-SINGLECARRIER45}},$$

where both packet orientations and the full Bézout-dependent reciprocal numerator remain active. The parallel  $b$ -diagonal node **BDIAGONAL-AFFINEA-SURVIVING-VERTEX-RECT45** is independently open. The earlier R7 uniform- $k = 0$ /one-level-Möbius residual is retained below as historical research provenance but is superseded as the controlling public first frontier.

The supplied Aristotle C0/unitary-Fourier bank records a successful 8298-job, zero-error kernel build and ordinary Mathlib axiom footprint. Its latest modules depend on a larger formal graph than is currently synchronized into public main, so formal and research status remain explicitly separated. Erdős #287 remains open.

## Status at a glance

| Item                                                            | Status                         | Meaning                                                               |
|-----------------------------------------------------------------|--------------------------------|-----------------------------------------------------------------------|
| Finite $M \leq 4 \times 10^9$ exclusion                         | KERNEL-PROVED / UNCHANGED      | Strongest unconditional public theorem.                               |
| C0 unitary-Fourier core                                         | ANALYTICALLY BANKED            | Hostile repairs included; exact four-term contraction.                |
| Exact product collision                                         | ANALYTICALLY BANKED            | Closed analytic core, not the whole problem.                          |
| Double Type II                                                  | ANALYTICALLY BANKED            | Closed analytic core, subject to the source architecture.             |
| C0                                                              | CONDITIONAL SOURCE PIN         | Analytically closed conditional on physical/formal normalisation.     |
| SHAREDGCD-OMEGA $\mathcal{H}$ -NORMALISATION-SOURCEPIN45        | OPEN SOURCE PIN                | Physical $\Omega_H$ and Perron/nuclear normalisation unless absorbed. |
| Naive transverse two-signless DFT                               | RETRACTED                      | Full CRT leaves moving numerator coefficients.                        |
| One-conductor reciprocity                                       | ANALYTICALLY BANKED            | Repaired transverse reciprocity normal form.                          |
| $q_C$ unitary Fourier                                           | ANALYTICALLY BANKED            | Arbitrary $L^2$ vectors; no Möbius cancellation required.             |
| Dense- $q_C$ compiler                                           | BANKED ON EXPLICIT HYPOTHESES  | Strict subregion only.                                                |
| Full transverse branch                                          | STRICT REDUCTION / OPEN        | Critical cells survive.                                               |
| THREEFACTOR-TRANSVERSE-CRITICAL-BEZOUTNUMERATOR-SINGLECARRIER45 | FIRST ANALYTIC RESIDUAL / OPEN | Both orientations and Bézout numerator remain.                        |
| BDIAGONAL-AFFINEA-SURVIVING-VERTEX-RECT45                       | PARALLEL OPEN                  | Kept separate from transverse.                                        |
| Supplied Aristotle C0 bank                                      | KERNEL-CHECKED BUNDLE          | 8298 jobs, zero errors; public dependency-complete import pending.    |
| Gate 0–2 / FCL / effectivity / WindowPairSupply                 | OPEN / DOWNSTREAM              | No implication is collapsed.                                          |
| Erdős Problem #287                                              | OPEN                           | No unconditional solution claim.                                      |

**Status vocabulary.** KERNEL-PROVED is reserved for statements checked by the Lean kernel in the stated formal bank. ANALYTICALLY BANKED means the research/paper theorem survived the current analytic audit; it is not automatically a Lean proof. CONDITIONAL SOURCE PIN exposes a literal physical-source obligation. STRICT REDUCTION means the parent has been reduced but not closed. OPEN, SUPERSEDED, and RETRACTED have their literal meanings. Historical material is retained for provenance even when it is no longer controlling.

## Contents

|                                                                            |          |
|----------------------------------------------------------------------------|----------|
| <b>Status at a glance</b>                                                  | <b>3</b> |
| <b>1 31 August 2026 C0/transverse synchronization</b>                      | <b>5</b> |
| 1.1 C0 exact product and Double Type II . . . . .                          | 6        |
| 1.2 C0 corrections . . . . .                                               | 6        |
| 1.3 Transverse repaired reciprocity and finite Fourier . . . . .           | 6        |
| 1.4 Exact current frontier . . . . .                                       | 6        |
| 1.5 Formal/Aristotle firewall . . . . .                                    | 7        |
| <b>2 The problem and the scope of this note</b>                            | <b>7</b> |
| <b>3 The machine-checked finite theorem</b>                                | <b>7</b> |
| 3.1 Holes and the adjacent-hole contradiction . . . . .                    | 8        |
| 3.2 Prime-power windows . . . . .                                          | 8        |
| 3.3 The interval-certificate chain . . . . .                               | 8        |
| 3.4 Sign-sensitive Sophie witnesses and the free maximum blocker . . . . . | 9        |
| <b>4 The exact large-<math>M</math> closure criterion</b>                  | <b>9</b> |

|           |                                                                                                |           |
|-----------|------------------------------------------------------------------------------------------------|-----------|
| <b>I</b>  | <b>Theory and Verification Manual: Gate 0–2</b>                                                | <b>10</b> |
| <b>5</b>  | <b>What the Gate theory is trying to do</b>                                                    | <b>10</b> |
| 5.1       | Verification vocabulary . . . . .                                                              | 11        |
| <b>6</b>  | <b>Gate 0: the physical sequence/comparison theorem</b>                                        | <b>11</b> |
| 6.1       | The physical #287 sequence . . . . .                                                           | 11        |
| 6.2       | Gate-0 target package . . . . .                                                                | 12        |
| <b>7</b>  | <b>Gate 1A: canonical common-weight Type-II provider</b>                                       | <b>13</b> |
| 7.1       | Parameters, rows and physical source . . . . .                                                 | 13        |
| 7.2       | Gate-1A target theorem . . . . .                                                               | 13        |
| 7.3       | The five load-bearing pins . . . . .                                                           | 14        |
| 7.4       | What the BPP mechanism proves and what it does not . . . . .                                   | 15        |
| 7.5       | The source-glue/SB- $\nu$ problem . . . . .                                                    | 15        |
| <b>8</b>  | <b>Gate 1B: signed/Möbius parity-breaking provider</b>                                         | <b>15</b> |
| 8.1       | Native centering, sign and complementary-divisor geometry . . . . .                            | 15        |
| 8.2       | The banked one-completion source . . . . .                                                     | 16        |
| 8.3       | The Motohashi line . . . . .                                                                   | 16        |
| 8.4       | Endpoint residue coefficient and diagonal/off-diagonal split . . . . .                         | 17        |
| 8.5       | Current Gate-1B research frontier: Bézout dual core . . . . .                                  | 17        |
| 8.6       | Downstream Gate-1B theorem objects . . . . .                                                   | 19        |
| <b>9</b>  | <b>Ford–Maynard generated packets and proof-specific Type II</b>                               | <b>19</b> |
| 9.1       | Published universal wrapper . . . . .                                                          | 19        |
| 9.2       | What their proof actually generates . . . . .                                                  | 19        |
| 9.3       | The three source obligations . . . . .                                                         | 20        |
| 9.4       | Provider-facing generated families . . . . .                                                   | 20        |
| <b>10</b> | <b>Balanced <math>R_9</math>, fixed-certificate leakage and FCL</b>                            | <b>21</b> |
| <b>11</b> | <b>Gate 2: source-specific Ford–Maynard lower-bound compiler</b>                               | <b>22</b> |
| <b>12</b> | <b>Current #287 specialization: Balanced7 bank and the uniform <math>k = 0</math> frontier</b> | <b>23</b> |
| 12.1      | Balanced7: current post-audit bank . . . . .                                                   | 23        |
| 12.2      | Uniform $k = 0$ : the next source parent . . . . .                                             | 26        |
| 12.3      | Retracting the old large-shared- $g_0$ and near-frequency global closures . . . . .            | 26        |
| 12.4      | Short-lift uniformity repair . . . . .                                                         | 26        |
| 12.5      | Product-modulus sign compression and fixed- $D$ rigidity . . . . .                             | 27        |
| 12.6      | CRT reciprocal normal form and the fixed- $n$ firewall . . . . .                               | 28        |
| 12.7      | Shared-gcd Gram-as-square: the main structural reduction . . . . .                             | 28        |
| 12.8      | Provisional one-level energy subpolytope . . . . .                                             | 29        |
| 12.9      | Why complete primitive- $t$ Ramanujan completion is nonclosing . . . . .                       | 29        |
| 12.10     | Current first exact main-line residual . . . . .                                               | 29        |
| <b>13</b> | <b>Verification manual: what is actually proved</b>                                            | <b>30</b> |
| 13.1      | Table 1: public and supplied machine-checked objects . . . . .                                 | 30        |
| 13.2      | Table 2: externally audited analytic inputs . . . . .                                          | 31        |
| 13.3      | Table 3: open source/provenance identifications . . . . .                                      | 31        |
| 13.4      | Table 4: current open analytic theorems . . . . .                                              | 31        |
| <b>14</b> | <b>Current numbered route to Erdős #287</b>                                                    | <b>31</b> |
| <b>15</b> | <b>How an external reviewer can falsify or improve the programme</b>                           | <b>32</b> |

|                                                                                       |               |
|---------------------------------------------------------------------------------------|---------------|
| <b>16 Glossary of recurring programme terms</b>                                       | <b>33</b>     |
| <b>17 The source-minimal affine algebra</b>                                           | <b>33</b>     |
| 17.1 The identity $\Lambda = \mu * \log$ . . . . .                                    | 34            |
| 17.2 Exact factor partition and determinant-one line . . . . .                        | 34            |
| <b>18 The balanced-seven factorial Euler reduction</b>                                | <b>34</b>     |
| <b>19 Published distribution results and the exact dictionary audit</b>               | <b>35</b>     |
| <b>20 The current end-to-end dependency graph</b>                                     | <b>36</b>     |
| <b>21 Formal verification and reproducibility</b>                                     | <b>37</b>     |
| 21.1 Public manuscript repository . . . . .                                           | 37            |
| 21.2 Later supplied safe bank . . . . .                                               | 38            |
| 21.3 Determinant-one formalization generations . . . . .                              | 38            |
| 21.4 External analytic firewall . . . . .                                             | 38            |
| <b>22 Conclusion</b>                                                                  | <b>38</b>     |
| <br><b>II Current Frontier and Provenance Ledger</b>                                  | <br><b>40</b> |
| <b>23 Purpose, coverage baseline and status firewall</b>                              | <b>40</b>     |
| <b>24 Gate 1A: current canonical-provider ledger</b>                                  | <b>40</b>     |
| 24.1 What is banked and what is still external . . . . .                              | 41            |
| 24.2 When Gate 1A is required for #287 . . . . .                                      | 41            |
| <b>25 Gate 1B: banked socket and current Bézout-dual frontier</b>                     | <b>42</b>     |
| 25.1 Downstream Gate-1B firewall . . . . .                                            | 43            |
| <b>26 Ford–Maynard generated packets: census before provider</b>                      | <b>43</b>     |
| 26.1 Current provider-facing census . . . . .                                         | 43            |
| <b>27 Balanced <math>R_9</math>, fixed-certificate leakage and the Gate-2 landing</b> | <b>44</b>     |
| 27.1 Gate 2 consumes, rather than invents, the packet bounds . . . . .                | 44            |
| <b>28 Current #287 frontier after Balanced7</b>                                       | <b>45</b>     |
| <b>29 Master current-status ledger</b>                                                | <b>46</b>     |
| <b>30 Superseded, false and provenance-only routes</b>                                | <b>47</b>     |
| <b>31 Formal theorem and provenance map</b>                                           | <b>47</b>     |
| <b>31 August 2026 controlling final status</b>                                        | <b>49</b>     |

## 1 31 August 2026 C0/transverse synchronization

This section is the controlling analytic checkpoint for this revision. It changes the frontier/status assignment, not the underlying Gate 0–2 theory manual. Earlier R7 uniform- $k = 0$  material remains later in the document as a dated research branch.

### 1.1 C0 exact product and Double Type II

After canonical reduced-projective conditioning the Kloosterman kernel becomes independent of the balanced  $b$  source. For  $b = \ell_b d_b e_b$  with  $d_b, e_b > Y$ , combine  $n = \ell_b d_b$ . Residue aggregation modulo  $x$  and inversion on unit residues give a restricted discrete Fourier matrix of exact operator norm  $\sqrt{x}$ . With fixed-depth physical product energy,

$$\eta_b^2 \ll L^C \left( \frac{x}{B} + \frac{1}{N} + \frac{1}{E_b} + \frac{1}{x} \right).$$

The product-energy theorem is asserted only for the literal bounded dyadic physical packet; the complete unit branch is  $\gcd(ba_\rho b_\rho u_\rho, x) = 1$ ; nonunit cells belong to D4; and small- $x$  cells route to the short-conductor/low- $Q$  owner. Constants are chosen in the order structure, packet losses,  $K_*$ ,  $K_{\text{tr}}$ ,  $(K_Y, K_x)$ , then  $k$ . Thus exact product collision and Double Type II are analytically closed, and C0 is analytically closed conditional on formal normalisation.

### 1.2 C0 corrections

The old post-conditioning index  $n = j\bar{u}' \pmod{x}$  and old wrap  $1 + UU'/B$  are **RETRACTED** as controlling C0 architecture. Historical occurrences are preserved. The full physical source is no longer advertised as requiring a moving-level inverse-convolution large sieve after canonical projective cancellation.

### 1.3 Transverse repaired reciprocity and finite Fourier

For a packet,  $R_P = z^2 q_C q_m$  with  $q_C = 2(e/a_1)(r_2/c_2)b_1^\flat$ , and symmetrically for the primed packet. Full CRT fusion of  $q_C$  and  $q_m$  leaves cross-modulus inverse coefficients in the numerator, so the naive two-signless DFT is **RETRACTED**.

For

$$\Phi_P = e_q(A\bar{r}\bar{m})e_r(B\bar{q}\bar{m}),$$

reciprocate only the  $q$  component. With

$$\Gamma_P = -A + mB\bar{m} \pmod{r}, \quad g_P = (B - A, r), \quad m_P = rm/g_P,$$

one obtains

$$\Phi_P = \text{Arch}_P e_{m_P}(\Gamma_P^{\text{red}} \bar{q}), \quad (\Gamma_P^{\text{red}}, m_P) = 1.$$

This is **THREEFACTOR-TRANSVERSE-ONECONDUCTOR-RECIPROCITY45**.

If  $q = S_1 S_2$ , the finite operator gives

$$\left| \sum \alpha(S_1) \beta(S_2) e_{m_P}(\Gamma_P^{\text{red}} \overline{S_1 S_2}) \right| \ll L^C \sqrt{m_P(1 + L_1/m_P)(1 + L_2/m_P)} \|\alpha\|_2 \|\beta\|_2,$$

and hence

$$\eta_P^2 \ll L^C \left( \frac{m_P}{L_1 L_2} + \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{m_P} \right).$$

This is **THREEFACTOR-TRANSVERSE-qC-UNITARYFOURIER45**. It accepts arbitrary  $L^2$  vectors and spends no Möbius cancellation. The dense- $q_C$  compiler closes only packets satisfying its explicit dominance/two-long-source-group hypotheses.

### 1.4 Exact current frontier

The older **THREEFACTOR-TRANSVERSE-CARRIERFACTORIZATION-PAIR-EXISTENCE45** is **SUPERSEDED AS FIRST FRONTIER**. The first current analytic residual is

$$\boxed{\text{THREEFACTOR-TRANSVERSE-CRITICAL-BEZOUTNUMERATOR-SINGLECARRIER45}}.$$

The surviving operator retains both packet orientations and the full Bézout-dependent reciprocal numerator. It includes nondominant  $q_C/q'_C$  cells, partitions with one short side, single atomic-carrier dominance, and critical  $q \asymp rm$  geometry. Candidate second Fourier axes include the Bézout numerator, an opposite-packet carrier, the  $m$  variable after  $TT^*$ , a signed  $L^2$  carrier, grouped short factors, or both orientations. None is promoted here.

The parallel branch

BDIAGONAL-AFFINEA-SURVIVING-VERTEX-RECT45 : OPEN

is kept separate.

### 1.5 Formal/Aristotle firewall

The supplied 31-August bundle records the completed C0 unitary-Fourier / transverse-carrier-interface bank and a default `lake build` of 8298 jobs with zero errors. New declaration axiom audits report only ordinary Mathlib foundations such as `propext`, `Classical.choice`, and `Quot.sound`, with no reported `sorry`, custom axiom, `unsafe`, opaque shortcut, `implemented_by`, or `native_decide`. The named newest files depend on a larger formal graph than the current public-main tree, so the paper records the verified supplied bank without claiming that public `RequestProject/Main.lean` already reproduces the 8298-job build.

## 2 The problem and the scope of this note

The public problem asks whether, for distinct integers

$$1 < n_1 < \cdots < n_k, \quad \sum_{i=1}^k \frac{1}{n_i} = 1,$$

one necessarily has

$$\max_{1 \leq i < k} (n_{i+1} - n_i) \geq 3.$$

It is listed as open in the Erdős Problems archive and originates in the sources cited there [1, 2].

For a finite set  $A \subset \mathbb{N}$ , write

$$N = \min A, \quad M = \max A.$$

A counterexample is a finite set satisfying

$$A \subset \{2, 3, \dots\}, \quad \sum_{a \in A} \frac{1}{a} = 1,$$

and the local gap condition that every nonmaximal  $a \in A$  is followed by  $a + 1$  or  $a + 2$  in  $A$ . The Lean development formalizes exactly this predicate and proves the ordered form from it.

This note is deliberately not a chronological research diary. It records only the current controlling finite theorem, the exact conditional closure interface, the source-minimal algebra, the factorial Euler balanced-seven reduction, and the current open analytic obligations. Earlier squarefree/collision routers, Vaughan decompositions, failed provider dictionaries, hostile audits, and superseded residuals belong to the separate technical verification ledger and immutable V15 archive.

## 3 The machine-checked finite theorem

**Theorem 3.1** (Finite exclusion). *There is no exact counterexample to Erdős Problem #287 with*

$$3 \leq M \leq 4,000,000,000.$$

*Equivalently, every unit-fraction representation in the problem with maximum denominator at most  $4 \times 10^9$  has a consecutive gap at least 3.*

The theorem is represented in the repository by `no_Erdos287Counterexample_of_max_le_4e9`. Its proof is not an exhaustive enumeration of finite subsets. It is a certificate compiler built from the three ideas below.

### 3.1 Holes and the adjacent-hole contradiction

For a hypothetical counterexample, call an integer  $x \in [N, M]$  a *hole* when  $x \notin A$ . The  $\text{gap} \leq 2$  condition implies that two consecutive integers in  $[N, M]$  cannot both be holes. This is the formal theorem `Gap2CE.holes_isolated`.

A second elementary fact places the top half of the interval above the minimum:

$$N \leq \left\lfloor \frac{M}{2} \right\rfloor.$$

Thus, whenever  $M \leq 2x$  and  $x + 1 \leq M$ , the consecutive pair  $x, x + 1$  lies inside the closed window where the adjacent-hole contradiction applies.

### 3.2 Prime-power windows

For  $j \geq 0$ , let  $C(j)$  denote the largest reduced numerator arising from a nonempty partial reciprocal sum

$$\sum_{s \in S} \frac{1}{s}, \quad \emptyset \neq S \subseteq \{1, \dots, j\}.$$

The formal development uses a certified upper table

$$C_{\text{val}}(0), \dots, C_{\text{val}}(9) = 1, 1, 3, 11, 25, 137, 137, 1019, 2143, 7129.$$

The prime-power window theorem says, schematically, that if  $p$  is prime,  $e \geq 1$ ,  $p^e \mid x$ , and

$$C\left(\left\lfloor \frac{M}{p^e} \right\rfloor\right) < p,$$

then  $x$  cannot belong to  $A$ . The reason is that the top  $p$ -adic layer of the reciprocal sum would force a nonzero numerator to vanish modulo  $p$ .

Consequently, if both  $x$  and  $x + 1$  possess such prime-power divisors, then they are two adjacent holes. This gives the central finite compiler.

**Proposition 3.2** (Window-pair blocker). *Suppose there are primes  $p_u, p_v$ , positive exponents  $a_u, a_v$ , and an integer  $x$  such that*

$$\begin{aligned} p_u^{a_u} \mid x, \quad p_v^{a_v} \mid x + 1, \\ \left\lfloor \frac{M}{p_u^{a_u}} \right\rfloor, \left\lfloor \frac{M}{p_v^{a_v}} \right\rfloor \leq 9, \\ C_{\text{val}}\left(\left\lfloor \frac{M}{p_u^{a_u}} \right\rfloor\right) < p_u, \quad C_{\text{val}}\left(\left\lfloor \frac{M}{p_v^{a_v}} \right\rfloor\right) < p_v, \end{aligned}$$

and

$$M \leq 2x, \quad x + 1 \leq M.$$

Then a  $\text{gap} \leq 2$  counterexample with maximum  $M$  cannot exist.

### 3.3 The interval-certificate chain

One certificate can exclude an entire interval  $L \leq M \leq U$ . The divisibility data are fixed at a chosen  $x$ , monotonicity controls the windows as  $M$  varies, and  $U \leq 2x$  keeps the forced holes above the minimum. The repository contains a chain of 34 certificates whose intervals meet end-to-end and cover

$$[3, 4 \times 10^9].$$

All primality side conditions are discharged in the kernel; the construction does not use `native_decide`.



### 3.4 Sign-sensitive Sophie witnesses and the free maximum blocker

The latest supplied finite bank sharpens the final analytic-to-finite landing and is worth stating because it prevents small endpoint mistakes later. A *plus* Sophie witness at  $M$  is a prime  $q$  with

$$q \geq 5, \quad M < 3q, \quad 2q + 1 \leq M, \quad 2q + 1 \text{ prime},$$

equivalently  $M/3 < q \leq (M-1)/2$  with the integer endpoints understood literally. A *minus* witness is

$$q \geq 5, \quad M < 3q, \quad 2q \leq M, \quad 2q - 1 \text{ prime},$$

so the correct minus endpoint is

$$q \leq \left\lfloor \frac{M}{2} \right\rfloor.$$

Either witness contradicts an exact counterexample. The hypothesis  $q \geq 5$  is not cosmetic: for a prime it is the same as  $q > 3$ , which is the finite window threshold needed by the cofactor-two obstruction.

There is also a free maximum-divisor blocker. If  $q > 3$  is prime,  $q \mid M$  and  $M < 3q$ , then necessarily  $M \in \{q, 2q\}$ ; in either case the top-layer/cofactor-two exclusion contradicts  $M \in A$ . Thus a counterexample cannot have prime maximum and cannot have  $M = 2q$  with  $q > 3$  prime. These are finite repairs, not analytic inputs; they should be used when translating positive affine mass into an actual blocker.

## 4 The exact large- $M$ closure criterion

**Definition 4.1** (Window-pair supply). For  $M \in \mathbb{N}$ ,  $\text{WindowPairSupply}(M)$  is the assertion that there exist  $x, p_u, a_u, p_v, a_v \in \mathbb{N}$  such that

$$\begin{aligned} p_u, p_v \text{ are prime,} \quad a_u, a_v \geq 1, \\ p_u^{a_u} \mid x, \quad p_v^{a_v} \mid x + 1, \\ \left\lfloor \frac{M}{p_u^{a_u}} \right\rfloor \leq 9, \quad C_{\text{val}}\left(\left\lfloor \frac{M}{p_u^{a_u}} \right\rfloor\right) < p_u, \\ \left\lfloor \frac{M}{p_v^{a_v}} \right\rfloor \leq 9, \quad C_{\text{val}}\left(\left\lfloor \frac{M}{p_v^{a_v}} \right\rfloor\right) < p_v, \\ M \leq 2x, \quad x + 1 \leq M. \end{aligned}$$

In plain terms, two consecutive integers in the top half of  $[1, M]$  each have a prime-power divisor larger than roughly  $M/10$ , with a small local numerator condition.

**Theorem 4.2** (End-to-end conditional compiler). *Assume there is an explicit  $M_0 \leq 4 \times 10^9$  such that*

$$\text{WindowPairSupply}(M) \quad \text{for every } M \geq M_0.$$

*Then Erdős Problem #287 is true.*

This implication is machine-checked as `no_Erdos287Counterexample_of_closure`. The proof splits at  $4 \times 10^9$ : the finite theorem handles the lower range, while the window-pair blocker handles every larger maximum.

The interface is weaker than a Sophie-Germain prime condition. For example, if a prime  $q$  in the interval  $M/3 < q \leq (M-1)/2$  has  $2q + 1$  prime, then one may take  $x = 2q$ . The analogous minus case uses  $x = 2q - 1$ . The Lean theorem `windowPairSupply_of_sophieWitness` formalizes this implication. The present programme does not claim such Sophie witnesses for every large  $M$ .

## Part I

# Theory and Verification Manual: Gate 0–2

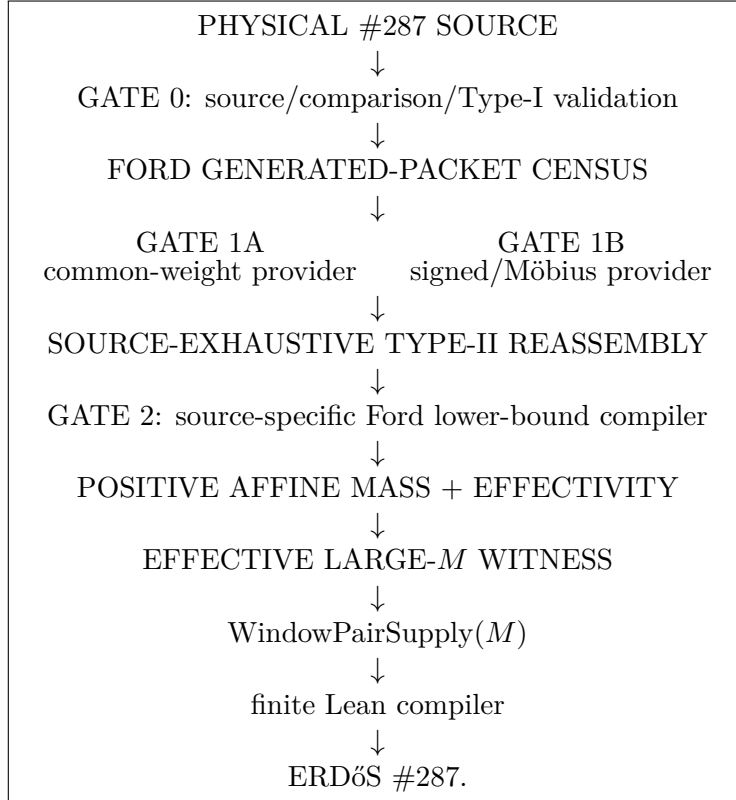
## 5 What the Gate theory is trying to do

This Part is written for a reader who has never seen the earlier chats, the V15 patch stack, or the repository. The Gate labels are not theorem substitutes: each Gate is a literal mathematical package with an input source, a target theorem, a verification boundary, and specified downstream consequences.

Conceptually, the programme separates four jobs.

- Gate 0.** Construct and validate the physical sequence/comparison class: prime mass, comparison regularity, growth, Type I, and the exact expected-term convention.
- Gate 1.** Supply parity-breaking Type-II estimates for the actual generated packets.
- Gate 1A.** One canonical common-weight moving-source provider.
- Gate 1B.** A complementary signed/Möbius/high-order provider that preserves the arithmetic sign resource through completion/dispersion.
- Gate 2.** Consume a source-exhaustive Type-II package, the Ford arithmetic weight, the exceptional  $N_2$  sector, and one compatible  $\varepsilon_*$  to run the Ford–Maynard lower-bound endgame.

The intended dependency graph is



(Theory-DAG)

This is a dependency graph, not a proof.

The following non-implications are load-bearing:

Gate 1A closed  $\nRightarrow$  Gate 1B closed,      Gate 1B closed  $\nRightarrow$  Gate 1A closed,  
 Gate 1A + Gate 1B  $\nRightarrow$  Full Type II without source exhaustiveness,  
 generated Type II  $\nRightarrow$  Gate 2 without Gate-0/ $N_2$ /source inputs,  
 Balanced7 closed  $\nRightarrow$  FCL closed,  
 FCL closed  $\nRightarrow$  WindowPairSupply( $M$ ),  
 none of the preceding statements alone proves Erdős #287.

(Theory-firewall)

## 5.1 Verification vocabulary

Every major object below is assigned one of the following meanings.

| Label                               | Meaning                                                                                        |
|-------------------------------------|------------------------------------------------------------------------------------------------|
| LEAN-PROVED                         | Proved by a cited/current Lean kernel replay.                                                  |
| PROVED-FINITE /<br>PROVED-ALGEBRAIC | Exact finite/combinatorial/algebraic mathematics; may or may not also be Lean formalized.      |
| EXTERNALLY-AUDITED                  | A sourced mathematical theorem or source audit, not formalized here.                           |
| CONDITIONAL-COMPILER                | The implication is proved, but one or more explicit antecedents remain open.                   |
| SOURCE-OPEN                         | The physical/source identification, normalization or dictionary is missing.                    |
| ANALYTIC-OPEN                       | The source object is sufficiently fixed; a deep estimate remains unproved.                     |
| SOURCE-BLOCKED                      | The literal source object/prefactor needed even to state the final theorem is not yet printed. |
| CAPACITY-ONLY                       | Exponent arithmetic passes, but no analytic theorem follows.                                   |
| FALSE / RETIRED                     | The literal proposed route is disproved or withdrawn.                                          |
| SUPERSEDED                          | Historically valid but no longer the controlling representation.                               |

## 6 Gate 0: the physical sequence/comparison theorem

### 6.1 The physical #287 sequence

Fix a nonnegative smooth weight  $W$  supported inside the analytic blocker band; the finite blocker itself only needs the corresponding interval to lie in the top half. Define

$$\mathfrak{S}_2 = \prod_{\ell > 2} \frac{\ell(\ell-2)}{(\ell-1)^2}, \quad B(n) = \mathfrak{S}_2 \prod_{\substack{\ell|n \\ \ell > 2}} \frac{\ell-1}{\ell-2},$$

and

$$a_X(n) = W(n/X)(\Lambda(2n-1) + \Lambda(2n+1)),$$

$$b_X(n) = 4W(n/X)B(n), \quad w_X(n) = a_X(n) - b_X(n). \quad (\text{G0-source})$$

The role of  $b_X$  is to model the expected local density of the two affine prime forms;  $w_X$  is the discrepancy on which Type I/II cancellation is sought.

The elementary comparison algebra includes

$$B(n) = \mathfrak{S}_2 \sum_{\substack{e|n \\ e \text{ odd squarefree}}} \prod_{\ell|e} \frac{1}{\ell-2},$$

and for each fixed  $d$ ,

$$\lim_{Y \rightarrow \infty} \frac{1}{Y} \sum_{m \leq Y} B(dm) = \frac{d}{\phi(2d)}. \quad (\text{G0-Bmean})$$

Thus one sign has mean  $2d/\phi(2d)$  and the two-sign model has mean  $4d/\phi(2d)$  at the fixed- $d$  level. Also

$$B(n) \ll \log \log(3n), \quad |w_X(n)| \ll \log X,$$

and, in the present source class,

$$\sum_{n \lesssim X} |w_X(n)| \tau(n) \ll X (\log X)^2. \quad (\text{G0-growth})$$

## 6.2 Gate-0 target package

**Definition 6.1** (Gate-0 package). Gate 0 is *closed for the #287 source at scale  $X$*  if the following seven items hold with one common physical normalization.

G0.1  $a_X(n) \geq 0$  and the comparison sequence  $b_X$  belongs to the Ford–Maynard ordinary sequence class used in the lower-bound theorem.

G0.2 Prime mass:

$$\sum_p b_X(p) \gg \frac{X}{\log X}.$$

G0.3 Growth/divisor control, including a bound of the form (G0-growth) (or the exact stronger divisor-weight norm required by the chosen Ford invocation).

G0.4 Comparison regularity: the physical  $b_X$  satisfies the Ford–Maynard condition (b.2) uniformly over every coagulation region actually used downstream. At the fixed- $d$  level this is compatible with (G0-Bmean); the required uniform publication-level statement must be source-pinned.

G0.5 Literal maximal/weighted Type I:

$$\left| \sum_{m \leq X^{1/2-\varepsilon}} \tau(m)^A \max_I \left| \sum_{\substack{X/2 < mn \leq X \\ n \in I}} w_X(mn) \right| \right| \ll_{A,W,\varepsilon} X (\log X)^{-A}. \quad (\text{G0-I})$$

G0.6 One fixed principal/small-conductor/exceptional-character expected-term convention is used consistently by both the prime source and the comparison source.

G0.7 Even, nonunit, prime-power and boundary strata created by the affine forms  $2n \pm 1$  are either included in the source theorem or routed with a proved error.

**Current verification.** Items G0.1–G0.3 have elementary/source-level proofs in the dossier; the exact finite problem statement and finite blocker are Lean-proved. G0.4 and G0.5 are **SOURCE-OPEN / EXTERNALLY-AUDITED TARGETS** at the exact publication normalization; G0.6 is the same physical comparison convention that later appears in the factorial balanced-seven and Pure5 comparison pins.

**What Gate 0 closing buys.** It supplies the non-parity hypotheses needed by the Ford lower-bound compiler and removes the possibility that a later Type-II estimate is being applied to the wrong comparison sequence.

**What it does not buy.** Gate 0 does not prove Type II, parity breaking, Gate 1A, Gate 1B, Gate 2, a large- $M$  witness, or Erdős #287.

## 7 Gate 1A: canonical common-weight Type-II provider

### 7.1 Parameters, rows and physical source

At an auxiliary scale  $X_1$ , set

$$\begin{aligned} M &= X_1^{1/3}, & R &= X_1^a, & L &= X_1^b, \\ H &= X_1^{a+2b-2/3}, & K &= \frac{M}{R}, & U &= \frac{L}{H}, & D &= \frac{L^2}{H}. \end{aligned} \quad (\text{G1A-scales})$$

Thus

$$RK = M, \quad DH = L^2, \quad M^2H = RL^2.$$

A row is

$$e = (r, k, m), \quad r \asymp R \text{ prime}, \quad 1 \leq |k| \leq K, \quad m \asymp M, \quad m' = m + kr \asymp M.$$

Choose the canonical affine forms  $A_e(t), B_e(t)$  so that

$$\boxed{m'A_e(t) - mB_e(t) = 2k.} \quad (\text{G1A-det})$$

For

$$\rho_\ell(y) = \mathbf{1}_{\ell|y} - \frac{1}{\ell}$$

and bounded prime coefficients  $b_p, d_q$ , the *one common physical-weight* source is

$$\boxed{C_e(b, d) = \sum_{t \in \mathbb{Z}} W_D(t) \left( \sum_{p \asymp L} b_p \rho_p(A_e(t)) \right) \left( \sum_{q \asymp L} d_q \rho_q(B_e(t)) \right).} \quad (\text{G1A-source})$$

The word “common” is literal:  $W_D$  is one datum. Arbitrary independently chosen weights  $W_{D,e}$  are a stronger source class and are not covered by the canonical theorem.

### 7.2 Gate-1A target theorem

**Theorem 7.1** (Canonical common-weight Gate-1A provider – conditional form). *Assume the five source/analytic pins G1A-A through G1A-E below. Then, uniformly for the canonical all- $m$  row family  $\mathcal{E}^\sharp$ ,*

$$\boxed{\sum_{e \in \mathcal{E}^\sharp} |C_e(b, d)|^2 \ll \frac{ML^4}{H} X_1^{o(1)} \left( \sum_{p \asymp L} |b_p|^2 \right) \left( \sum_{q \asymp L} |d_q|^2 \right).} \quad (\text{G1A-target})$$

*The exact normalized variant used by the formal compiler is equivalent after the banked  $H^2$  normalization bridge.*

The theorem is a **CONDITIONAL-COMPILER**: the finite/operator part is substantially banked, but the physical source pins are not all inhabited.

### 7.3 The five load-bearing pins

**G1A-A: Poisson–Bruhat/source transport.** After the one-sided physical transform and reciprocity, every retained source state  $\xi$  must have moving-prime coefficient

$$c_r(\xi) = B_\xi F_\xi(r/R), \quad (\text{G1A-PB})$$

where  $F_\xi$  is a uniformly smooth envelope generated from the one physical  $W_D$ , not an arbitrary row weight. The transport must preserve the physical/normalized energy up to a square-summable source error.

STATUS: SOURCE-OPEN / CONDITIONAL PIN.

Without this pin, the later moving-prime participation argument would apply to a convenient model rather than to the physical packet.

**G1A-B: absolute source ceiling.** The transported physical source must satisfy

$$\mathcal{T}_{\text{abs}} \ll M^2 L^4 X_1^{o(1)}. \quad (\text{G1A-abs})$$

This is the absolute comparator against which the moving-family contraction is measured.

STATUS: SOURCE/ANALYTIC PIN.

**G1A-C: fixed-state exclusion and prime participation.** A fixed nonprojective state must not absorb the moving prime variable. Smoothness of  $F_\xi$  produces a long plateau; the short-interval prime theorem then gives

$$N_\xi \gg R^{3/4-o(1)} \quad (\text{G1A-part})$$

participating prime rows for each retained state. The finite BPP compiler converts this participation into

$$E_{\text{np}} \ll R^{-1/2+o(1)} \mathcal{T}_{\text{abs}}. \quad (\text{G1A-np})$$

The project kernel-checks the finite participation-to-energy implication; it does not formalize the entire physical smooth-envelope/short-interval input.

**Status.** FINITE COMPILER LEAN-PROVED; the physical participation statement is EXTERNAL / SOURCE-OPEN.

**G1A-D: outer-root physical energy and projective routing.** The abstract nonprojective energy must be identified with the physical Gate energy, and the projective/axis/diagonal branch must be routed in the same source normalization. The finite ratio-class algebra includes the state diagonal in the zero-curvature branch, and the physical projective mass target is

$$E_{\text{proj}} \ll MHL^4 X_1^{o(1)}. \quad (\text{G1A-proj})$$

Exactly one outer square root converts (G1A-np) to an  $R^{-1/4+o(1)}$  contraction. At the three controlling vertices the signed margins are

$$\frac{1}{72}, \quad \frac{1}{24}, \quad \frac{1}{32}. \quad (\text{G1A-margins})$$

**Status.** FINITE SQUARE-ROOT / PROJECTIVE ALGEBRA BANKED; the physical energy identification is SOURCE-OPEN.

**G1A-E: recombination/discarded-sector bound.** After all source truncations, exceptional routes and recombinations, the physical error must satisfy

$$E_{\text{rec}} \ll U^{-2+o(1)} M^2 L^4, \quad (\text{G1A-rec})$$

with the source-exact prefactor and without spending the same saving twice. The exponent margins at the controlling vertices are

$$\frac{1}{18}, \quad \frac{1}{18}, \quad \frac{1}{24}.$$

**Status.** EXPONENT BUDGET BANKED; literal source recombination is SOURCE-OPEN.

## 7.4 What the BPP mechanism proves and what it does not

The finite BPP mechanism is a local-to-family compiler:

$$\begin{array}{c} \text{smooth moving-prime participation} \\ \Downarrow \\ R^{-1/2+o(1)} \text{ family-energy contraction} \\ \Downarrow \\ R^{-1/4+o(1)} \text{ after one outer root.} \end{array}$$

It does *not* prove the source transport, absolute source ceiling, physical energy identification or recombination estimate. Those are separate statements precisely so that a reader can attack or falsify them independently.

## 7.5 The source-glue/SB- $\nu$ problem

The current source-glue question is whether the same physical packet can simultaneously satisfy the common-weight representation, the BPP state representation and the final routing representation with fibre multiplicity

$$N_\nu = X_1^{o(1)} \quad (\text{G1A-SBnu})$$

in the synchronized two-view model. This is a source-identification frontier rather than a new exponent computation.

Finally,

$$\text{GATE1A\_REQUIRED\_FOR\_287} = \text{UNKNOWN.}$$

The Ford-generated packet census, not programme nomenclature, decides whether a needed #287 packet truly belongs to the canonical common-weight source class.

# 8 Gate 1B: signed/Möbius parity-breaking provider

## 8.1 Native centering, sign and complementary-divisor geometry

For coprime  $d, p$ ,

$$\rho_{dp}(y) = \rho_d(y)\rho_p(y) + \frac{\rho_d(y)}{p} + \frac{\rho_p(y)}{d}. \quad (\text{G1B-center})$$

If  $q$  is squarefree and  $p \mid q$  is the distinguished prime, then

$$\mu(q/p) = -\mu(q). \quad (\text{G1B-sign})$$

This is the sign resource that must not be squared away accidentally.

The native complementary-divisor source exposes

$$\boxed{q\ell - uv = 2.} \quad (\text{G1B-det})$$

This determinant-two geometry is separate from the Gate-1A determinant  $2k$  and from the #287  $\mu * \log$  determinant-one line.

## 8.2 The banked one-completion source

On dyadic windows  $d \sim D$  and  $p \sim P$ , the authoritative one-completion coefficient is

$$\boxed{\beta_{D,P}(d, p) = \mu_D(d) \Lambda_P(p),} \quad (\text{G1B-beta})$$

where  $\mu_D, \Lambda_P$  are the physical truncated Möbius and prime weights on the stated windows. The public formal challenge proves the separate linearity/bilinearity of this coefficient and an exact level-of-distribution budget requiring

$$\boxed{\theta \geq \frac{1}{2} + \frac{1}{104}.} \quad (\text{G1B-104})$$

It also proves the implication from an inhabited source-to-convolution-BV interface to the abstract Gate conclusion. Producing that interface from the literal physical source remains open. This is a valid formal provider socket, but it is no longer the first research-level analytic obstruction.

## 8.3 The Motohashi line

Fix  $u, \ell$  and a base solution to

$$\boxed{uv_0 + 2 = \ell z_0.} \quad (\text{G1B-base})$$

All solutions on the complementary-divisor line are

$$\boxed{v_t = v_0 + \ell t, \quad z_t = z_0 + ut.} \quad (\text{G1B-line})$$

Let  $H$  be the physical line length and write  $e_H(x) = e^{2\pi i x/H}$ . For the exact one-completion coefficient and a smooth cutoff  $W_1$  define

$$\boxed{\mathcal{Z}_{u,\ell}(k) = \sum_t \beta_{D,P}(z_t) W_1(t/H) e_H(-kt).} \quad (\text{G1B-Z})$$

For the five centred defect factors

$$b_5 = \delta_1 * \delta_2 * \delta_3 * \delta_4 * \delta_5$$

and a second physical cutoff  $W_2$ , define

$$\boxed{\mathcal{B}_{u,\ell}(k) = \sum_t b_5(v_t) W_2(t/H) e_H(-kt).} \quad (\text{G1B-B})$$

The Motohashi pairing is

$$\boxed{\mathcal{M}_{u,\ell} = \frac{1}{H} \sum_k \mathcal{Z}_{u,\ell}(k) \overline{\mathcal{B}_{u,\ell}(k)}.} \quad (\text{G1B-M})$$

The current supplied research ledger reports the exact A/B/C source pin, the twisted-defect A/B/C reduction and the fivefold Motohashi iteration as passed source algebra; the fixed-polylogarithmic- $k$  interior is closed modulo the comparison convention.



## 8.4 Endpoint residue coefficient and diagonal/off-diagonal split

At the endpoint define, for a unit residue  $a \bmod \ell$ ,

$$A_{\ell,a,k} = \sum_{u \equiv -2a^{-1} \pmod{\ell}} a_4(u) \mathcal{Z}_{u,\ell}(k), \quad (\text{G1B-A})$$

where  $a_4$  is the four-defect coefficient inherited from the physical source. The endpoint Cauchy energy is

$$\mathcal{E}_{\text{end}}(k) = \sum_{\ell \sim L} \sum_{a \in (\mathbb{Z}/\ell\mathbb{Z})^\times} |A_{\ell,a,k}|^2. \quad (\text{G1B-Eend})$$

Opening the square gives

$$\mathcal{E}_{\text{end}}(k) = \sum_{\ell,a} \sum_{u_1, u_2 \equiv -2a^{-1} \pmod{\ell}} a_4(u_1) \overline{a_4(u_2)} \mathcal{Z}_{u_1,\ell}(k) \overline{\mathcal{Z}_{u_2,\ell}(k)}.$$

The diagonal is  $u_1 = u_2$ . The off-diagonal is parametrized by

$$u_2 = u_1 + j\ell, \quad j \neq 0. \quad (\text{G1B-j})$$

For corresponding line parameters  $t_1, t_2$ ,

$$z_2(t_2) - z_1(t_1) = u(t_2 - t_1) + j v_{t_2}, \quad (\text{G1B-off-source})$$

and, after opening  $\beta = \mu_D \Lambda_P$ ,

$$d_2 p_2 - d_1 p_1 = u(t_2 - t_1) + j v_{t_2}. \quad (\text{G1B-off-beta})$$

These identities are the literal arithmetic object behind the current endpoint off-diagonal problem.

The latest supplied research ledger records

$$\text{endpoint } u\text{-diagonal} = \text{power closed with } X^{-1/12+o(1)}. \quad (\text{G1B-diag-status})$$

## 8.5 Current Gate-1B research frontier: Bézout dual core

The previous repair label **SHIFTED-MAM-TOPBAND45** and the earlier endpoint label **RANKONE-ENDPOINT-U-OFFDIAG45** remain useful provenance, but neither is the current first exact Gate-1B analytic residual. The later centred-AP/Bézout reduction strictly reduces the parent **C4SHIFT-OFFDIAG-CENTERED-AP5/8-GRAM45** to

$$\boxed{\text{C4SHIFT-BEZOUT-DUAL-ROELL45}}.$$

Its status is **ANALYTIC OPEN / FIRST EXACT GATE-1B RESIDUAL AT RESEARCH LEVEL**; the quantitative  $D$ -census promotion feeding this node remains **HOSTILE-AUDIT PENDING**.

On the clean sector

$$(d, \ell) = 1, \quad r = dr_0,$$

the physical Bézout line is

$$\boxed{bZ_2 - aZ_1 = \ell r_0},$$

with parametrization

$$Z_1 = \ell r_0 \kappa_{a,b} + bt, \quad Z_2 = \ell r_0 \bar{b}_a + at.$$

The physical AP residue is

$$\boxed{t \equiv -2\overline{usdab} \pmod{\ell}},$$

and is independent of  $r_0$ . After cyclic dualization the centred kernel is

$$\widehat{K}_{\text{cent}}(\nu) = \frac{e_\ell(-\nu\tau)}{\ell} \mathfrak{q}_{\ell, \xi_1, \xi_2, dab}(\nu),$$

where

$$\mathfrak{q}(\nu) = 1 - \frac{1}{\ell} \sum_{k \bmod \ell} \mathcal{M}_1(k + \nu \overline{dab}) \overline{\mathcal{M}_2(k)}.$$

The zero dual mode is a *local owner*, not zero; it belongs to the parallel source comparison child

$$\boxed{\text{C4SHIFT-BEZOUT-ZEROMODE-LOCALMATCH45: \quad SOURCE \quad OPEN}},$$

itself inside TOPBAND-BROAD-MAJOR-TREE-MATCH45.

For nonzero dual frequency the analytic phase is

$$e_\ell\left(2\nu\overline{usdab}\right).$$

Opening the two-coordinate gamma source gives

$$bx_2y_2 - ax_1y_1 = \ell r_0, \quad e_\ell(\nu t) = e_\ell(\nu \bar{b} x_1 y_1).$$

The latest research ledger proposes the  $D$ -local estimate

$$\mathcal{N}_D \ll Y^{3/4} \sqrt{\min(D, T/D)} L^C, \quad T = Y^{3/2} L^{O(1)}.$$

This closes the boundary  $D$ -bands provisionally, but the central range remains. At the worst balanced point  $D = T^{1/2} = Y^{3/4}$  the provisional bound leaves residual

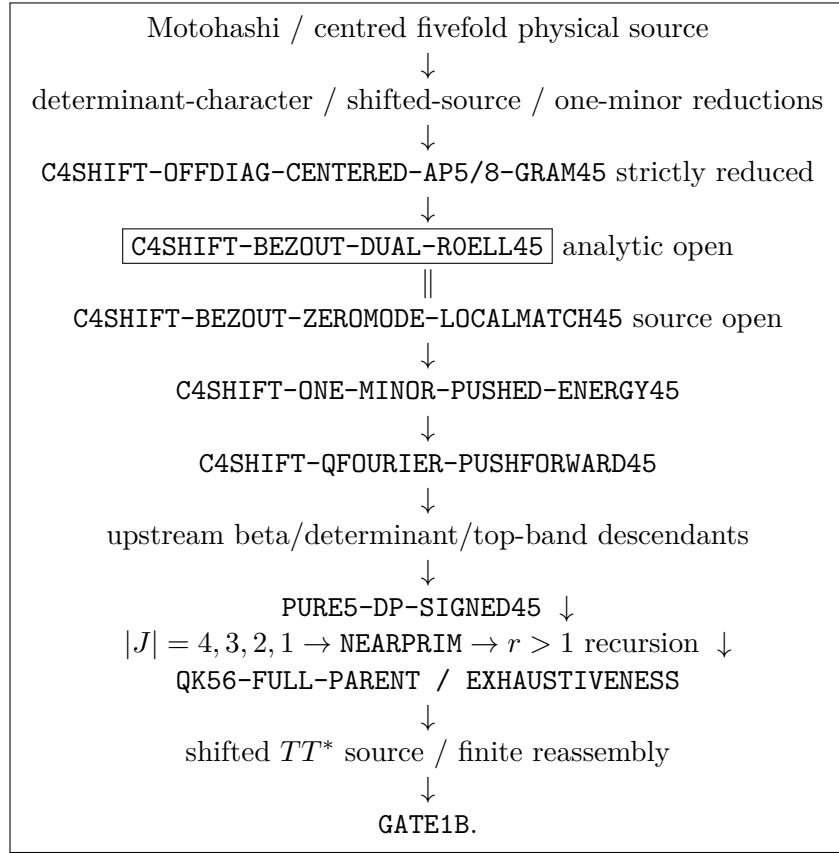
$$Y^{3/8} = X^{1/24}.$$

The fixed- $(d, a, b, \ell)$  uniqueness/census used to reach this statement must still be hostile-audited with the possibility that two admissible  $t$  values differ by a multiple of  $\ell$ . No literal Kloosterman/spectral provider is currently matched.

Thus the missing Gate-1B mechanism is joint cancellation of the nonzero  $t$ -dual reciprocal phase against the exact shifted gamma Bézout source while retaining the coupled  $(\nu, r_0, \ell)$  family. Gate 1B remains open.

## 8.6 Downstream Gate-1B theorem objects

The current Gate-1B firewall is



The zero dual mode/local-match branch may not be skipped. The provisional  $D$ -local bound is not promoted until its census passes hostile audit, and no downstream node is closed merely by naming the Bézout dual residual.

## 9 Ford–Maynard generated packets and proof-specific Type II

### 9.1 Published universal wrapper

The Ford–Maynard lower-bound framework uses a universal Type-II hypothesis. In the present 1/6 specialization a safe sufficient form is

$$\left| \sum_{\substack{(X/2)^\varepsilon < m \leq X^{1/6-\varepsilon} \\ X/2 < mn \leq X}} \xi_m \kappa_n w_X(mn) \right| \ll_A X (\log X)^{-A} \quad (\text{FM-universal-II})$$

for every allowed divisor-bounded coefficient pair. The project does *not* claim that Ford–Maynard publish the weaker proof-specific theorem described next.

### 9.2 What their proof actually generates

The Ford arithmetic source has

$$G(m; n) = \mathbf{1}_{m \leq n^\gamma} \lambda(m_1) g(\mathbf{v}(m_2; n)), \quad H(n) = \sum_{d|n} G(d; n), \quad (\text{FM-GH})$$

with the canonical rough/smooth split  $m = m_1 m_2$ . After finite decomposition and condition removal, Proposition 7.22 reaches a fixed-depth product sum (equation (7.23) in the source)

$$\sum_{n=n_1 \cdots n_N \asymp x} w_n x_1(n_1) \cdots x_N(n_N), \quad (\text{FM-7.23})$$

and chooses a product  $\prod_{e \in E} n_e$  in the Type-II range. Grouping the selected factors and their complement gives

$$\boxed{\sum_{n=n' n'' \asymp x} Y_1(n') Y_2(n'') w_{n' n''}} \quad (\text{FM-generated-II})$$

The published proof then invokes the universal Type-II hypothesis.

Hence an invocation-by-invocation replacement theorem would be enough for this proof:

|                                                                                                                                                                  |                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| every actual generated (FM-generated-II) packet<br>has a proved source-exact provider<br>$\Downarrow$<br>every Type-II invocation in Proposition 7.22 is covered | (FM-replacement-logic) |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|

This implication additionally requires a source-exhaustive census and a replacement certificate; it is not obtained merely by identifying some packets.

### 9.3 The three source obligations

**Definition 9.1** (SOURCE-ADAPTER45). For every generated packet, print the literal operator, physical ranges, coefficient class, weight dependence, centering/expected term, gcd/collision restrictions, original prefactor and normalized target, and map it to a proved provider without changing the source class.

**Definition 9.2** (FM-TO-GATE-COORDINATE-CENSUS45). Enumerate every Type-II packet actually generated by the Ford decomposition, with enough coordinate information to decide whether it belongs to Gate 1A, Gate 1B or a direct exceptional/standard class.

**Definition 9.3** ( $C_{\text{FM}}$  replacement certificate). Verify invocation by invocation that the generated census covers every use of the universal Type-II hypothesis in the chosen Ford proof, with no omitted packet, no multiplicity loss and no double-spending of an analytic saving.

All three remain open source/reassembly obligations.

### 9.4 Provider-facing generated families

| Family           | Literal source/scale                                                                                | Candidate provider                  | Why it matches / what is missing                                                                                                                            |
|------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Möbius singleton | one selected<br>$m < X^{1/6}$ fac-<br>tor carries Möbius<br>weight; complement<br>remains generated | Gate 1B signed<br>source            | Möbius sign is in the cor-<br>rect physical slot only after a<br>literal completion dictionary;<br>generated outer-factor stabil-<br>ity is still required. |
| Model singleton  | selected factor is<br>model/Mellin-like;<br>Möbius carrier<br>may remain in the<br>complement       | QK56 or direct<br>affine completion | provider cannot be inferred<br>from the word “singleton”; na-<br>tive packet dictionary is open.                                                            |

| Family                         | Literal source/scale                                                            | Candidate provider                                               | Why it matches / what is missing                                                                                       |
|--------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Balanced-seven V-cell          | seven fixed prime boxes at $Y = X^{1/7}$                                        | factorial- $\Omega_7$ source, then the fixed-degree-seven attack | factorial Euler encoding is exact; endpoint/3221 analytic theorem and comparison remain.                               |
| Pure5/high-order family        | five centred defect factors with distinguished-prime/Möbius source              | Gate 1B Motohashi/rank-one chain                                 | current first analytic object is the endpoint off-diagonal; high- $k$ and downstream defect/conductor families remain. |
| Balanced $R_9$                 | nine-coordinate Ford-generated leakage cell                                     | fixed-certificate/FCL lane                                       | finite binomial 70 is proved; literal physical $G(d; n)$ specialization is source-open.                                |
| Common-weight packets          | actual generated source must preserve one physical $W_D$ after scale conversion | Gate 1A                                                          | requires exact scale/source dictionary and all five Gate-1A pins.                                                      |
| Remaining fixed- $g_*$ packets | not yet exhaustively censused                                                   | TBD from literal source                                          | the finite census/provider assignment itself is still open.                                                            |

## 10 Balanced $R_9$ , fixed-certificate leakage and FCL

The balanced order-nine finite identity is

$$\sum_{j=0}^4 (-1)^j \binom{9}{j} = 70. \quad (\text{R9-70})$$

This is PROVED-FINITE; by itself it is not the physical Ford statement  $H(n) = 70$ .

To obtain  $H(n) = 70$  one must prove that the literal Ford specialization of  $G(d; n)$  on the balanced  $R_9$  cell selects exactly the corresponding low/high divisor orders and has the required value of  $g(\emptyset)$ . That is

$$\boxed{\text{literal } G(d; n) \text{ specialization: SOURCE-OPEN.}} \quad (\text{R9-G-open})$$

Therefore balanced  $R_9$  is a *Ford-generated leakage candidate*, not a completed leakage theorem.

For one fixed positive Ford certificate  $g_*$ , define its arithmetic weight

$$H_*(n) = \sum_{d|n} G_*(d; n).$$

Let  $\mathcal{P}$  be the desired prime region,  $\mathcal{N}_1$  the certificate-sign composite region,  $\mathcal{N}_2$  the exceptional collar, and

$$\mathcal{U}_* = (X/2, X] \setminus (\mathcal{P} \cup \mathcal{N}_1 \cup \mathcal{N}_2).$$

The fixed-certificate leakage is

$$\boxed{\mathcal{L}_*(X) = \sum_{n \in \mathcal{U}_*} w_X(n) H_*(n).} \quad (\text{FCL})$$

The  $N_2$  error is

$$\mathcal{E}_{N_2}(X) = \sum_{n \in \mathcal{N}_2} w_X(n) H_*(n).$$

If the total correlation, leakage,  $N_2$  error and comparison error are all small relative to the positive certificate margin  $C_*(\varepsilon_*)$ , then the fixed-certificate transference gives positive affine prime mass. A sufficient quantitative form is

$$|\mathcal{L}_*(X)| + |\mathcal{E}_{N_2}(X)| \leq \frac{C_*(\varepsilon_*)}{4} B_X, \quad (\text{FCL-margin})$$

together with the corresponding total-correlation/comparison bounds. This is a conditional compiler; it does not itself prove FCL.

Balanced  $R_9$  may contribute to  $\mathcal{L}_*$  once the source obligation (R9-G-open) is discharged, but closing that one cell does not close the full leakage family.

## 11 Gate 2: source-specific Ford–Maynard lower-bound compiler

Fix one compatible

$$P_{\varepsilon_*} = \left( \frac{1}{2} - \varepsilon_*, \varepsilon_*, \frac{1}{6} - 2\varepsilon_* \right).$$

Gate 2 is the theorem-shaped endgame package:

**Theorem 11.1** (Gate-2 lower-bound compiler – conditional). *Assume:*

G2.1 *the Gate-0 source/comparison/Type-I package;*

G2.2 *the literal Ford  $G(d;n)/H(n)$  source and its sign properties;*

G2.3 *a source-specific aggregate  $N_2$  estimate;*

G2.4 *one common fixed  $\varepsilon_* > 0$  compatible with every source range;*

G2.5 *either the universal Type-II theorem (FM-universal-II), or a complete generated-packet provider package together with **SOURCE-ADAPTER45**, **FM-TO-GATE-COORDINATE-CENSUS45** and  $C_{\text{FM}}$ ;*

G2.6 *the ordinary/bounded-class firewall and every comparison/exceptional-character normalization needed by the chosen Ford theorem;*

G2.7 *effective versions of all inputs at a computable threshold.*

*Then the Ford–Maynard lower-bound geometry yields positive affine mass of the form*

$$\boxed{\sum_{p \in \mathcal{P}} a_X(p) \geq (C_*(\varepsilon_*) + o(1)) B_X \gg \frac{X}{\log X}}, \quad (\text{G2-positive})$$

*for the physical #287 source and the chosen certificate.*

STATUS: CONDITIONAL-COMPILER / NOT ACTIVATED.

The theorem still does not by itself produce **WindowPairSupply**. One additional effective witness extraction theorem must convert the positive affine mass into a Sophie/safe-prime or prime-power blocker configuration, or directly into the **WindowPairSupply**( $M$ ) predicate of the finite compiler. That final conversion/effectivity step remains open.

## 12 Current #287 specialization: Balanced7 bank and the uniform $k = 0$ frontier

### 12.1 Balanced7: current post-audit bank

The controlling fixed-certificate source is the SP-2 smooth-parity packet

$$G_*(m; n) = \mathbf{1}_{m \leq n^{\gamma*}} \mu(m_1) g_*(\mathbf{v}(m_2; n)), \quad H_*(n) = \sum_{m|n} G_*(m; n).$$

On the balanced seven-prime cell

$$P = p_1 \cdots p_7, \quad p_i \asymp X^{1/7},$$

the source coefficient is

$$H_*(P) = 1 - 7 + 21 - 35 = -20.$$

The literal one-sign packet is

$$-20 \sum_{\mathbf{p}} \Omega_{\lambda}(\mathbf{p}) [\Lambda(2P + s) - 2B(P)], \quad s = \pm 1.$$

The source-minimal factorization

$$\Lambda(2P + s) = \sum_{qr=2P+s} \mu(q) \log r$$

forces the physical logarithmic source ledger

$$C_{\text{ext}} = 1;$$

the earlier  $C_{\text{ext}} = 0$  claim is superseded.

**Full- $q$  Euler principal owner.** Define

$$J_P(z) = \sum_{\substack{q \leq z \\ (q, 2P)=1}} \frac{\mu(q)}{\phi(q)} \log \frac{z}{q}.$$

The audited Euler factorization is

$$F_P(w) = \frac{H_P(w)}{\zeta(1+w)}, \quad H_P(0) = 2B(P),$$

with  $H_P(w) = O(1)$  uniformly on the required contour for the seven-prime family, so

$$J_P(z) = 2B(P) + O_A((\log z)^{-A})$$

uniformly for  $z = 2P \pm 1$ . This is one full- $q$  principal identity, not a separate main term for each dyadic  $q$  cell.

**Exact SmallQ/SmallR/Hard partition.** With  $U = X^{1/3}$ , split  $qr = 2P + s$  into

$$q \leq U, \quad q > U, \quad r \leq U, \quad q > U, \quad r > U.$$

The sectors are exhaustive and disjoint at the sharp level; the Hard sector is subsequently dyadically smoothed.

**SmallQ hostile-audited pass.** The former direct Gate-0 Type-I dictionary is superseded as a controlling provider because  $q$  divides  $2P + s$ , not  $P$ . The replacement is a primitive-conductor split  $3 + 4$  multiplicative large sieve. The literal three- and four-prime coefficients satisfy

$$\sum |a_3|^2 \ll X^{3/7}(\log X)^{-3+o(1)}, \quad \sum |b_4|^2 \ll X^{4/7}(\log X)^{-4+o(1)}.$$

For squarefree  $q \asymp Q$ , the conductor split has the form

$$L_{Q,D}(c) \ll (\log X)^{O(1)} \left( Q + \frac{N}{D} \right) \|c\|_2^2,$$

because  $q = fk \asymp Q$ ,  $f > D$  implies  $k \ll Q/D$ . The low primitive-conductor sector with  $D = (\log X)^{30}$  is routed through Siegel–Walfisz; the high-conductor sector also passes. Hence

AFFINE287-SP2-SMALLQ-34LS-NORMALIZATION45: HOSTILE-AUDITED PASS.

**SmallR owner and general-modulus conductor split.** For arbitrary modulus  $r$ , if  $\chi$  is induced from primitive  $\chi^* \bmod f$  and  $j = r/f$ , the exact relation is

$$\chi_r(n) = \chi^*(n) \mathbf{1}_{(n,j)=1},$$

with no assumption that  $r$  is squarefree or  $(f, j) = 1$ . Also

$$\phi(fj) \geq \phi(f)\phi(j), \quad f > D, \quad fj \asymp R \implies j \ll R/D,$$

so the general-modulus conductor estimate

$$L_{\text{gen}}(R, D; c) \ll (\log X)^{O(1)} \left( R + \frac{N}{D} \right) \|c\|_2^2$$

survives.

Let

$$D_{\text{sr},s} = S_{\text{sr},s} - M_{\text{sr},s}^{\text{prin}}.$$

The principal restriction is owned by the one global Euler packet and satisfies

$$M_{\text{sr},s}^{\text{prin}} \ll X^{1/3+o(1)},$$

while the audited direct SmallR source obeys

$$S_{\text{sr},s} \ll X(\log X)^{-16.5+o(1)}.$$

Thus

$$D_{\text{sr},s} = o(X/\log X)$$

and

AFFINE287-SP2-SMALLR-OWNER-SUBTRACTION45: HOSTILE-AUDITED PASS.

**Hard physical range.** For

$$q \asymp Q = X^\theta, \quad \frac{1}{3} < \theta < \frac{2}{3}, \quad M = X^{2/7}, \quad W_5 = X^{5/7},$$

put

$$T = \frac{W_5}{Q} = X^{5/7-\theta}, \quad z = T^{1/20}.$$

The physical fixed margin is

$$\delta = \frac{5}{7} - \frac{2}{3} = \frac{1}{21}, \quad \log z \geq \frac{1}{420} \log X.$$



For primes  $\ell < z$ , if  $\ell \nmid q$  there is one forbidden  $t$  class modulo  $\ell$ ; if  $\ell \mid q$ , there is no forbidden class because  $\ell < z < Y$  cannot divide the physical  $w'$ . Hence the short- $t$  upper sieve gives

$$\#\{t \text{ admissible}\} \ll \frac{T}{\log X} \frac{q}{\phi(q)}.$$

The corresponding Shiu progression map has modulus exponent  $5/7$  below the admissible  $3/4$  threshold by

$$\frac{3}{4} - \frac{5}{7} = \frac{1}{28}.$$

The corrected variance ledger is

$$V_Q^{RR} \ll \frac{MW_5^2}{Q} (\log X)^{-5+o(1)},$$

not  $(\log X)^{-10}$ ; the second copy is consumed by the  $t$ -sieve. Moreover

$$V_Q^{PP} \ll \frac{MW_5^2}{Q} (\log X)^{-10+o(1)}, \quad V_Q^{RP} \ll \frac{MW_5^2}{Q} (\log X)^{-15/2+o(1)}.$$

Thus the raw-raw channel dominates:

$$V_Q \ll \frac{MW_5^2}{Q} (\log X)^{-5+o(1)}.$$

With the literal source cost  $C_{\text{ext}} = 1$ ,

$$|D_{Q,s}| \ll X (\log X)^{-5/2+o(1)},$$

and summing  $O(\log X)$  hard cells gives

$$\sum_Q |D_{Q,s}| \ll X (\log X)^{-3/2+o(1)} = o(X/\log X).$$

Hence

$$\text{HARD-THETA-PHYSICAL-DELTA45: HOSTILE-AUDITED PASS, } \delta = 1/21.$$

**Exact full- $q$  ownership and Balanced7 status.** Use exactly one global Euler principal packet:

| packet           | owner                                    |
|------------------|------------------------------------------|
| SmallQ principal | full Euler principal                     |
| SmallQ defect    | 3 + 4 conductor-split large sieve        |
| SmallR principal | full Euler principal                     |
| SmallR defect    | direct SmallR estimate                   |
| Hard principal   | full Euler principal                     |
| Hard defect      | short- $t$ /Shiu physical-range estimate |
| even/nonunit     | zero or explicit router                  |

with  $q = X^{1/3}$  owned by SmallQ and  $r = X^{1/3}$  by SmallR. Even  $q$  is impossible for  $2P + s$  odd; nonunit pieces are routed from the exact affine source. Therefore the principal restrictions sum to one and only one Euler principal packet:

$$\text{ALL-Q-NO-DOUBLE-SPENDING45: HOSTILE-AUDITED PASS},$$

$$\text{ALL-Q-PROVIDER-REASSEMBLY45: HOSTILE-AUDITED PASS}.$$

At the current analytic research/paper level,

$$\text{SP2-BALANCED7-FULL-Q45: PASS}$$

BALANCED7: HOSTILE-AUDITED RESEARCH/PAPER PASS].

The firewall is source provenance: the completed audit treats the literal SP-2 source,  $H_*(P) = -20$ , and seven prime-box source as supplied. The public manuscript repository does not presently expose the later 8213-job source-seal bank. Therefore this note states the Balanced7 verdict as conditional on the exact literal SP-2 source seal rather than claiming public-Lean source completeness.

**Balanced7 effectivity remains open.** The SmallQ low-conductor argument uses classical Siegel–Walfisz for  $f \leq (\log X)^{30}$ , whose threshold is generally ineffective. A separate lane remains

287-EFFECTIVE-LOWCOND-EXCEPTIONAL-PNT45: OPEN],

requiring an effective exceptional-zero-aware PNT/AP statement matching the project’s principal and exceptional comparison convention. No computable  $M_0$  is claimed.

## 12.2 Uniform $k = 0$ : the next source parent

Balanced7 does not complete the  $k = 0$  fixed-certificate source. The next parent is

287-K0-SP2-UNIFORM-FRAGMENTATION-REASSEMBLY45: OPEN].

It must handle the actual  $k$ -range, coefficients

$$H_k = (-1)^r \binom{k-1}{r}, \quad r = \lfloor k\gamma_* \rfloor,$$

uniform analytic constants, fragmentation/polytopal multiplicity, squareful and repeated-prime strata, model-containing states and an exhaustive unique-owner map.

Earlier source work proposed a three-small-prime-prefix child. The latest determinant-one audit has moved the exact main-line residual further and more structurally.

## 12.3 Retracting the old large-shared- $g_0$ and near-frequency global closures

The local reciprocal calculations remain valid:

SHAREDGO-BPAIR-AVERAGED45: PASS,      SHAREDGO-CAUCHY-CONFIGURATION45: PASS.

At Gram level the near-density gain is density<sup>1/2</sup> and after the final outer square root it is density<sup>1/4</sup>. However the former global claims

DET1-LARGESHAREDGO-CELLS45: CLOSED,      DET1-PRIMITIVE-NEARFREQ45: CLOSED

are retracted. Restoring the cross-level pair multiplicity of order  $G^2/H$ , the normalization  $1/(g_1 g_2)$ , the first friable Cauchy  $A$ -factor and the reciprocal energies leaves a signed cross-level problem. Thus fixed-pair cancellation does not by itself close the full family.

## 12.4 Short-lift uniformity repair

For

$$m_{g,b}(\Delta) = \sum_{(d,2bg)=1} \frac{\mu(d)}{d} \Psi(d/\Delta),$$

the finite Euler factor is

$$H_{\mathfrak{H}}(w) = \prod_{p|\mathfrak{H}} (1 - p^{-1-w})^{-1}.$$

The corrected contour is

$$\Re w = -\frac{c_0}{\sqrt{\log \Delta}},$$

not the earlier  $t$ -dependent contour. For

$$\Delta \geq D_{\sharp} = \exp(C_{\sharp}(\log \log X)^2)$$

and  $\mathfrak{H} \leq X^C$ , the extremal prime-divisor argument gives

$$H_{\mathfrak{H}}(w) \leq (\log X)^{o(1)}.$$

Hence  $\exp(-c\sqrt{\log \Delta})$  defeats any fixed power of  $\log X$  after choosing  $C_{\sharp}$  large. The current research status is

DET1-SHORTLIFT-EULER-COLLAPSE45: PASS AFTER UNIFORMITY REPAIR.

The analytic zero-free-region input remains uninhabited in Lean; no formal zeta zero-free theorem is inferred.

## 12.5 Product-modulus sign compression and fixed- $D$ rigidity

Write

$$g_1 = g_0 r_1, \quad g_2 = g_0 r_2, \quad (r_1, r_2) = 1, \quad (g_0, r_1 r_2) = 1,$$

in the squarefree clean sector, and set

$$n = r_1 r_2.$$

Then exactly

$$\boxed{\mu(g_1)\mu(g_2) = \mu(r_1)\mu(r_2) = \mu(n),} \quad [g_1, g_2] = g_0 n.$$

Thus two level Möbius signs compress to one. If

$$D = t_1 r_2 - t_2 r_1,$$

then  $(D, n) = 1$  in the primitive-frequency source, and for fixed  $D$

$$t_1 \equiv D \bar{r}_2 \pmod{r_1}, \quad t_2 \equiv -D \bar{r}_1 \pmod{r_2}.$$

The common residual freedom is the shared- $g_0$  parameter, while

$$\boxed{\frac{t_1}{g_1} - \frac{t_2}{g_2} = \frac{D}{g_0 n}.$$

Consequently the friable kernel is

$$\Phi_A\left(\frac{D}{g_0 n}\right),$$

independent of the divisor split  $r_1 \mid n$ .

The exact research labels are

DET1-LEVELPAIR-PRODUCTMOD-SIGN45: PASS

DET1-FIXEDD-FREQUENCY-RIGIDITY45: PASS.

These are latest research-exact reductions; their formal banking is pending in the public manuscript repository.

## 12.6 CRT reciprocal normal form and the fixed- $n$ firewall

After separating the shared- $g_0$  component, CRT combines the residual reciprocal phases modulo  $r$  and  $n/r$ . If

$$\beta_r \equiv b_1 \pmod{r}, \quad \beta_r \equiv b_2 \pmod{n/r},$$

the remaining phase has the exact form

$$e_n(sD \overline{2g_0\beta_r}).$$

Thus

$$\boxed{\text{DET1-LEVELPAIR-N-RECIPROCAL-NORMALFORM45: PASS}}$$

and the divisor-split kernel is a squarefree unitary-divisor additive-reciprocal/Kloosterman-idempotent object, not an ordinary Dirichlet character.

For fixed squarefree  $n$ , each prime  $p \mid n$  has exactly two states: assigned to  $r$  or  $n/r$ . After Mellin separation of balanced support, the divisor split factors over  $p \mid n$ . But the local factor contains  $\overline{n/p} \pmod{p}$ ; it therefore depends on the complementary coprime component. Hence

$$\boxed{\text{DET1-LEVELPAIR-PRIMEASSIGNMENT45: PASS AT FIXED } n},$$

but

$$\boxed{\text{DET1-LEVELPAIR-SINGLEMOBIUS-N45: OPEN}},$$

and no ordinary Euler-product/ $1/\zeta$  cancellation follows from the single  $\mu(n)$  compression.

## 12.7 Shared-gcd Gram-as-square: the main structural reduction

Let  $\Omega_H((g_1, g_2))$  select the shared-gcd  $H$  sector and define its divisor-lattice Möbius transform

$$\lambda_H(d) = \sum_{e \mid d} \mu(d/e) \Omega_H(e) = (\mu * \Omega_H)(d).$$

Exact Möbius inversion gives

$$\Omega_H((g_1, g_2)) = \sum_{\substack{d \mid g_1 \\ d \mid g_2}} \lambda_H(d).$$

Consequently the signed cross-level Gram reorganizes exactly as

$$Q_H = \sum_d \lambda_H(d) \sum_a \rho(a) \left| \sum_{d \mid g} \frac{\mu(g)}{g} V_g(a) \right|^2.$$

In the squarefree clean sector write  $g = dm$ ,  $(d, m) = 1$ . Since  $\mu(g) = \mu(d)\mu(m)$  and the outer  $\mu(d)$  disappears in the square,

$$Q_H = \sum_d \frac{\lambda_H(d)}{d^2} \sum_a \rho(a) \left| \sum_{\substack{m \\ (m,d)=1}} \frac{\mu(m)}{m} V_{dm}(a) \right|^2.$$

At research level the harmonic projector satisfies

$$\sum_d \frac{|\lambda_H(d)|}{d^2} \ll \frac{1}{H}$$

on the stated  $H$  support. Thus the former two-level  $G^2/H$  signed-pair combinatorics is replaced by harmonic shared-gcd mass  $1/H$  and a one-dimensional signed  $m$  family.

This is the principal new exact structural result:

$$\boxed{\text{DET1-SHAREDGCD-GRAM-SQUARE45: PASS EXACT}},$$

with public formal banking still pending.

## 12.8 Provisional one-level energy subpolytope

A coefficient-blind calculation gives the research bound

$$|Q_H| \ll \frac{(A+G)(GB+B^2)}{H} (\log X)^{o(1)}.$$

Against the required scale

$$AB^2(\log X)^{-6-2\eta-C_{\text{route}}},$$

it gives fixed-power saving on the strict subpolytope

$$\kappa > \max(\alpha, \theta) + \max(\beta, \theta) - 1 + \delta,$$

where

$$A = X^\alpha, \quad B = X^\beta, \quad G = X^\theta, \quad H = X^\kappa, \quad \alpha + \beta = 1.$$

The status is deliberately conservative:

$$\boxed{\text{DET1-SHAREDGCD-ONELEVEL-ENERGY45} \\ \text{PROVISIONAL RESEARCH CLOSURE ON A STRICT SUBPOLYTOPE}}$$

A hostile promotion audit is still required; this is not a publication theorem and does not alter the global OPEN status.

## 12.9 Why complete primitive- $t$ Ramanujan completion is nonclosing

Fully completing the primitive-frequency variable gives schematically

$$V_g(a) = \sum_b \beta_g(b) c_g(2ab + s).$$

For squarefree  $m$  and  $N = 2ab + s$ ,

$$p \nmid N : \quad c_p(N) = -1, \quad \mu(p) c_p(N) = +1,$$

while

$$p \mid N : \quad c_p(N) = p - 1, \quad \mu(p) c_p(N) = -(p - 1).$$

Thus the generic Ramanujan local sign cancels the remaining Möbius sign. The formal Euler object contains

$$\prod_p (1 + p^{-1-w}) = \frac{\zeta(1+w)}{\zeta(2+2w)},$$

which has a pole at  $w = 0$ , not a  $1/\zeta$  zero. Therefore

$$\boxed{\text{ONELEVEL-RAMANUJAN-ZEROFREE45: REPRESENTATION LOOP / NONCLOSING}}.$$

The remaining sign must be used before complete primitive-frequency Ramanujan reassembly neutralizes it. This is a source-preservation issue, not a generic fixed-Chowla obstruction.

## 12.10 Current first exact main-line residual

For the current shared-gcd sector define the one-level object

$$\mathcal{G}_{H,d} = \sum_{a \asymp A} \rho(a) \left| \sum_{\substack{m \asymp G/d \\ (m,d)=1}} \frac{\mu(m)}{m} \sum_{t \bmod dm}^* \tilde{G}_{dm,s,\tau}(t) e_{dm}(ta) \right|^2.$$

The required estimate is

$$\left| \sum_d \frac{\lambda_H(d)}{d^2} \mathcal{G}_{H,d} \right| \ll AB^2 (\log X)^{-6-2\eta-C_{\text{route}}}.$$

The current generic bound is only

$$\frac{(A+G)(GB+B^2)}{H} (\log X)^{o(1)}.$$

Therefore the first exact main-line analytic residual is

$$287\text{-K0-SP2-DET1-SHAREDGCD-ONELEVEL-MOBIUS-GRAM45}.$$

The missing mechanism is cancellation in

$$\sum_m \frac{\mu(m)}{m} V_{dm}(a)$$

while preserving the primitive  $t$  variable, reciprocal  $b$  source, friable  $a$  average and exact shared-gcd projector. Complete primitive- $t$  Ramanujan completion is explicitly excluded as the cancellation mechanism because it neutralizes the sign.

**Conceptual lesson.** The successive reductions expose a recurring source-preservation phenomenon. Local reciprocal cancellation is insufficient after the full cross-level multiplicity is restored. The successful reorganization instead keeps the Möbius signs until the shared-gcd Gram is converted exactly into a one-level signed operator. Conversely, complete primitive-frequency Ramanujan reassembly neutralizes the remaining Möbius sign. The present frontier is therefore a source-specific one-level Möbius–primitive-frequency dispersion problem, not a generic two-point Chowla problem.

## 13 Verification manual: what is actually proved

### 13.1 Table 1: public and supplied machine-checked objects

| Object                                                          | Verification status                               | Exact scope / nonclaim                                                                                                                                                                                       |
|-----------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Finite #287 theorem through $4 \times 10^9$                     | PUBLIC LEAN-CHECKED                               | Exact finite exclusion only; no large- $M$ theorem.                                                                                                                                                          |
| WindowPairSupply compiler                                       | PUBLIC LEAN-CHECKED / CONDITIONAL                 | Compiler from eventual effective supply; supply itself unproved.                                                                                                                                             |
| V15 $\mu * \log$ / determinant-one algebra                      | PUBLIC-MAIN SYNCHRONIZED                          | Finite/source algebra; no analytic cancellation.                                                                                                                                                             |
| V16 factorial Euler / local Euler / linearity                   | PUBLIC-MAIN SYNCHRONIZED                          | Factorial polarization and formal Euler algebra; no analytic endpoint theorem.                                                                                                                               |
| V17–V19 3221 finite/source banks                                | PUBLIC-MAIN SYNCHRONIZED                          | Grouping, finite energy, source reconstruction, pre-Poisson and character-Gram infrastructure; analytic inputs remain interfaces.                                                                            |
| Primitive Ramanujan / short-lift / shared- $g_0$ finite routers | SUPPLIED FORMAL BANK                              | Finite algebra/reassembly and local counting only; analytic zero-free inputs are uninhabited.                                                                                                                |
| Post-Balanced7 8213-job safe bank                               | SUPPLIED LEAN SAFE BANK / REPOSITORY SYNC PENDING | Reported modules include seven-box weights, conductor-split LS compiler, small-prime prefix, owner and uniform-fragmentation compilers; current public manuscript repository does not contain these modules. |
| Product-modulus / fixed- $D$ / CRT / Gram-square layer          | RESEARCH-EXACT / FORMAL BANKING PENDING           | Do not call public-Lean checked until a successful bank is supplied and synchronized.                                                                                                                        |

### 13.2 Table 2: externally audited analytic inputs

| Input                                              | Status                                     | Scope                                                                                        |
|----------------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------|
| Balanced7 SmallQ 3 + 4 conductor-split large sieve | HOSTILE-AUDITED PASS                       | Includes the literal $L^2$ norms and low/high primitive-conductor split.                     |
| General-modulus SmallR conductor large sieve       | HOSTILE-AUDITED PASS                       | Uses $\chi_r = \chi^* \mathbf{1}_{(n,j)=1}$ with no squarefree/coprime cofactor assumption.  |
| Short- $t$ one-dimensional upper sieve             | HOSTILE-AUDITED PASS                       | Physical hard range only; standard source-specific upper-sieve application.                  |
| Shiu divisor average                               | HOSTILE-AUDITED PASS                       | Progression modulus exponent 5/7 has margin 1/28 below the audited 3/4 threshold.            |
| Balanced7 full- $q$ analytic reassembly            | HOSTILE-AUDITED RESEARCH/PAPER PASS        | External analytic verdict; conditional on exact SP-2 source seal, not a Lean analytic proof. |
| Short-lift zero-free-region collapse               | RESEARCH/NANC PASS AFTER UNIFORMITY REPAIR | The zero-free analytic statement is not Lean-formalized.                                     |

### 13.3 Table 3: open source/provenance identifications

| Source obligation                                                 | Status                                | Meaning                                                                                                                                   |
|-------------------------------------------------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Exact SP-2 seven-box source seal                                  | SOURCE-OPEN IN PUBLIC MANUSCRIPT REPO | The hostile Balanced7 audit treats the physical seven-box source as supplied; later 8213-job formal bank is not public-synchronized here. |
| Gate-1A five physical pins / SB- $\nu$ source glue                | SOURCE-OPEN                           | Required only for packets actually censused into the canonical common-weight provider.                                                    |
| Gate-1B comparison/main-term and repaired top-band dictionary     | SOURCE/ANALYTIC OPEN                  | Independent from the present #287 uniform- $k = 0$ frontier.                                                                              |
| Ford generated-(7.23) census / SOURCE-ADAPTER45 / $C_{\text{FM}}$ | OPEN                                  | Needed only for proof-specific replacements that actually use the Ford generated family.                                                  |
| Balanced $R_9$ literal $G(d; n)$ specialization                   | SOURCE-OPEN                           | Finite 70 identity does not by itself give physical $H(n) = 70$ .                                                                         |
| Effective low-conductor exceptional PNT dictionary                | OPEN                                  | Needed for computable threshold/effectivity; classical Siegel–Walfisz is insufficient.                                                    |

### 13.4 Table 4: current open analytic theorems

| Theorem / node                                   | Exact task                                                                                                                                                                    | What remains if proved                                                                                                   |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 287-K0-SP2-DET1-SHAREDGCD-ONELEVEL-MOBIUS-GRAM45 | Obtain the required logarithmic cancellation in the shared-gcd one-level signed Möbius Gram while retaining primitive $t$ , reciprocal $b$ , friable $a$ and exact projector. | Uniform $k = 0$ reassembly/census, remaining fixed- $g_*$ packets, FCL, $N_2$ , Gate 2, effectivity and eventual supply. |
| DET1-SHAREDGCD-ONELEVEL-ENERGY45                 | Promote or refute the provisional strict-subpolytope coefficient-blind closure.                                                                                               | Even a positive audit does not automatically cover the full polytope.                                                    |
| Gate-1B<br>C4SHIFT-BEZOUT-DUAL-ROELL45           | Prove source-specific cancellation in the coupled nonzero $(\nu, r_0, \ell)$ Bézout/gamma dual family; audit the feeding $D$ -census.                                         | Zero-mode local match, upstream C4Shift promotion, Pure5/defect strata, QK56 and Gate 1B remain.                         |
| FCL / source-specific $N_2$                      | Control all remaining fixed-certificate leakage and exceptional collars.                                                                                                      | Gate 2 and effectivity.                                                                                                  |
| 287-EFFECTIVE-LOWCOND-EXCEPTIONAL-PNT45          | Replace ineffective low-conductor input with an effective exceptional-zero-aware theorem matching the comparison convention.                                                  | Effective Gate 2 and eventual WindowPairSupply still require the remaining source programme.                             |
| Eventual WindowPairSupply                        | Produce the effective adjacent prime-power-window witness for all sufficiently large $M$ .                                                                                    | Existing finite Lean compiler then proves the public statement.                                                          |

## 14 Current numbered route to Erdős #287

The current dependency route is:

1. source-seal the exact SP-2 Balanced7 seven-box adapter in the public formal record;

2. retain the hostile-audited Balanced7 research/paper closure, while keeping its effective version separate;
3. close 287-K0-SP2-DET1-SHAREDGCD-ONELEVEL-MOBIUS-GRAM45;
4. complete 287-K0-SP2-UNIFORM-FRAGMENTATION-REASSEMBLY45, including all actual  $k$  values, multiplicities, repeated/squareful strata and unique owners;
5. complete the remaining fixed- $g_*$  packet census and census-driven provider assignment;
6. close fixed-certificate leakage (FCL);
7. close the source-specific  $N_2$  collar;
8. activate Gate 2 at one compatible  $\varepsilon_*$ ;
9. discharge the separate effective low-conductor/exceptional PNT lane and make positive witness extraction computable;
10. prove eventual WindowPairSupply, or a stronger effective witness implying it;
11. invoke the existing finite Lean compiler.

Later results may collapse several steps. In particular, the first analytic residual above is a statement about the current determinant-one shared-gcd sector; it is not claimed to be an irreducible obstacle of every possible formulation.

The non-implications remain:

Balanced7 research closure  $\nRightarrow$  uniform  $k = 0$ ,  
 uniform  $k = 0 \nRightarrow$  FCL without packet exhaustiveness,  
 FCL  $\nRightarrow$  effective Gate 2 or WindowPairSupply,  
 no local packet estimate alone proves Erdős #287.

## 15 How an external reviewer can falsify or improve the programme

A serious review is most useful if it attacks one of the following:

1. a source dictionary that changes the physical packet;
2. a theorem-slot mismatch between an actual generated packet and a claimed provider;
3. a missing normalization or prefactor;
4. a multiplicity or support-enlargement loss;
5. double-spending of a Möbius sign or analytic saving;
6. an incorrect comparison/principal/exceptional-character main term;
7. an incomplete Ford-generated packet census;
8. an invalid invocation-by-invocation replacement of Ford–Maynard’s universal Type II;
9. an endpoint/effectivity gap;
10. a mismatch between a claimed Lean status and the actual kernel/public bank.

Failure of any conditional provider does *not* invalidate the machine-checked finite theorem through  $4 \times 10^9$ ; it invalidates only the corresponding large- $M$  route.



## 16 Glossary of recurring programme terms

|                               |                                                                                                                                                |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Gate 0</b>                 | The physical #287 source/comparison/Type-I package.                                                                                            |
| <b>Gate 1A</b>                | Canonical common-weight moving-source Type-II provider (G1A-source).                                                                           |
| <b>Gate 1B</b>                | Complementary signed/Möbius/high-order Type-II provider based on (G1B-center)–(G1B-M).                                                         |
| <b>Gate 2</b>                 | Source-specific Ford–Maynard lower-bound compiler.                                                                                             |
| <b>BPP</b>                    | Band-limited prime-participation mechanism converting moving-prime participation to a family-energy saving in Gate 1A.                         |
| <b>SB-<math>\nu</math></b>    | Synchronized two-view/source-glue fibre condition in the Gate-1A source adapter.                                                               |
| <b>Pure5</b>                  | Five-centred-defect distinguished-prime source family in the Gate-1B chain.                                                                    |
| <b>NearPrim</b>               | Near-primitive/full-conductor slice ( $r = 1$ ) of the Gate-1B conductor family.                                                               |
| <b>QK56</b>                   | Full same-/cross-modulus covariance parent (QK56-parent).                                                                                      |
| <b>SOURCE-ADAPTER45</b>       | Literal source dictionary from actual Ford-generated packets to proved providers.                                                              |
| $C_{\text{FM}}$               | Invocation-by-invocation certificate replacing the universal Type-II hypothesis in the chosen Ford proof by generated-packet estimates.        |
| $R_9$                         | Balanced nine-coordinate Ford-generated leakage candidate; finite binomial coefficient 70 is proved, physical $G(d; n)$ specialization is not. |
| <b>FCL</b>                    | Fixed-certificate leakage/correlation package (FCL).                                                                                           |
| $N_2$                         | Exceptional Ford collar requiring a separate source-specific aggregate bound.                                                                  |
| <b>Balanced7</b>              | Fixed-degree seven-prime-box V-cell at $Y = X^{1/7}$ .                                                                                         |
| <b>V-cell</b>                 | A generated packet in the model/factorial seven-prime branch.                                                                                  |
| <b>Möbius singleton</b>       | Generated short Type-II factor carrying the Möbius sign.                                                                                       |
| <b>fixed-<math>g_*</math></b> | One fixed positive Ford certificate used in the leakage transference.                                                                          |
| <b>WindowPairSupply</b>       | The exact large- $M$ adjacent prime-power window predicate consumed by the finite Lean compiler.                                               |

## 17 The source-minimal affine algebra

The current clean narrative starts from the exact identity  $\Lambda = \mu * \log$ . The supplied V15 Aristotle replay report records a successful 8118-job kernel build of the declarations summarized below. Because public repository **main** has not yet synchronized that replay, their status in this candidate is **LEAN-PROVED IN LATEST V15 REPLAY**, not **PUBLIC LEAN-CHECKED**. No analytic estimate is encoded by these identities.

### 17.1 The identity $\Lambda = \mu * \log$

For the genuine arithmetic functions,

$$\mu * \log = \mu * (\Lambda * 1) = (\mu * 1) * \Lambda = \Lambda,$$

and hence

$$\Lambda(N) = \sum_{qr=N} \mu(q) \log r.$$

At the affine value  $N = 2mn + s$ ,  $s \in \{-1, +1\}$ ,

$$\Lambda(2mn + s) = \sum_{qr=2mn+s} \mu(q) \log r.$$

The V15 replay also preserves the natural-number sign firewall for the minus case.

### 17.2 Exact factor partition and determinant-one line

For a cutoff  $U$ , factor pairs split disjointly into

$$q \leq U, \quad q > U, r \leq U, \quad q > U, r > U.$$

The hard class is exact rather than heuristic:

$$\sum_{\substack{qr=2mn+s \\ q>U, r>U}} \mu(q) \log r.$$

Every hard pair satisfies

$$qr - 2mn = s, \quad s = \pm 1.$$

This implies

$$(r, 2m) = 1, \quad (q, 2n) = 1.$$

Fixing  $r, m$  and one solution  $(q_0, n_0)$ , all integer solutions are

$$q = q_0 + 2mt, \quad n = n_0 + rt, \quad t \in \mathbb{Z},$$

with unique  $t$ . The V15 replay report records nonvanishing, the coprimality statements, both directions of this parametrization, and multiplicity-one uniqueness as kernel-checked.

The same replay proves only the finite exponent ledger, not an asymptotic estimate. In particular,

$$\frac{1}{6} - \frac{16623}{100000} = \frac{131}{300000} > 0.$$

## 18 The balanced-seven factorial Euler reduction

The factorial Euler family remains the clean algebraic encoding:

$$a_{\mathbf{z}}(p) = \frac{1}{7} \sum_{i=1}^7 z_i \omega_i(p), \quad F_{\mathbf{z}}(p^e) = \frac{a_{\mathbf{z}}(p)^e}{e!}.$$

For  $\Omega(n) = 7$ ,

$$7^7 [z_1 \cdots z_7] F_{\mathbf{z}}(n) = \sum_{\substack{p_1 \cdots p_7 = n \\ p_i \text{ ordered}}} \prod_{i=1}^7 \omega_i(p_i),$$

including repeated primes exactly. Coefficient extraction commutes with every linear expected-term operator.

The controlling physical source is nevertheless the fixed-certificate SP-2 packet, not a putative unique Ford-(7.23) generated packet. The completed hostile audit gives the research/paper-level Balanced7 full- $q$  pass described in Section 12: SmallQ, SmallR and Hard defects are controlled with one global Euler principal owner. The public-formal source seal of the literal seven-box SP-2 adapter is still pending, and the effective low-conductor version remains open.

Therefore the old sequence of first residuals

$$\begin{aligned} &3221\text{-DI-KUZNETSOV-LITERAL-SPLICE45,} \\ &3221\text{-HIGHQUOTIENT-FIVEBOX-SHIFTED-GRAM45,} \\ &\text{BALANCED7-OMEGA-FM723-SOURCE-ADAPTER45} \end{aligned}$$

is provenance only. The current main line has moved to uniform  $k = 0$  and the one-level Möbius Gram of Section 12.

## 19 Published distribution results and the exact dictionary audit

Drappeau–Granville–Shao work with the class  $\mathcal{C}$  of *multiplicative* functions satisfying  $|\Lambda_f(n)| \leq \Lambda(n)$ ; complete multiplicativity is only a special case. Their theorem reaches moduli  $x^{3/5-\varepsilon}$  for sufficiently smooth support, so it does not literally supply the exact endpoint  $Q = X^{3/5}$  required above.

For Pascadi’s current arXiv v2, the relevant stated result is the smooth-number theorem (Theorem 1.5, refined as Theorem 6.4). Immediately after Theorem 1.5 the paper remarks that a similar result for smooth-supported multiplicative functions can be deduced by a slight extension of Proposition 6.3; the current v2 does not state a separate “Theorem 1.7” giving the black box required here. More importantly, the written proof architecture itself is quantitatively nonclosing at the seven-prime smoothness scale.

Indeed, renaming the proof’s internal small parameter as  $\eta$ , the proof arranges

$$Q \leq x^{5/8-100\eta}, \quad y \leq x^\eta.$$

At the required modulus

$$Q = x^{3/5},$$

the first inequality implies

$$\frac{3}{5} \leq \frac{5}{8} - 100\eta \implies \boxed{\eta \leq \frac{1}{4000}}.$$

But the source has

$$y = x^{1/7},$$

so the second inequality would require

$$\boxed{\eta \geq \frac{1}{7}}.$$

These conditions are incompatible. Thus the *direct existing Pascadi proof route*, with its written parameters, does not furnish the factorial endpoint theorem at  $(Q, y) = (X^{3/5}, X^{1/7})$ . This is a theorem-dictionary/proof-parameter audit, not a claim that no future extension of Pascadi’s technology can succeed.

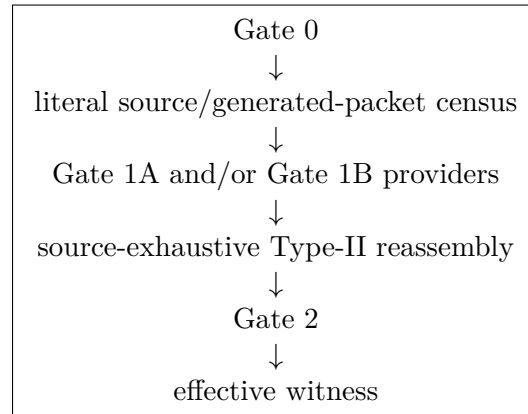
This explains the current research direction: preserve the literal fixed degree seven rather than forcing a generic smooth-factorization parameter, carry that structure through dispersion, and test a post-dispersion four-block arrangement. The “fixed-degree-seven / four-block” programme is a research direction, not a proved theorem or an already matched Gate-1B provider. Its purpose is precisely to attack the narrower signed torus coefficient without paying for a theorem uniform over all smooth-supported functions.

The earlier V14 Vaughan decomposition remains mathematically valid as a nonminimal alternative source audit. It is no longer presented as a co-equal controlling branch in this public narrative; its prime-modulus two-outer packet and related failed dictionaries belong in the technical provenance appendix.

**Current source-preservation lesson.** The earlier smooth-number, DI/Kuznetsov, HHH and large-shared- $g_0$  lanes remain valuable audit history, but none is the controlling uniform- $k = 0$  frontier. The latest exact reorganization keeps the Möbius sign through the shared-gcd Gram-as-square reduction. Fully completing the primitive frequency is explicitly nonclosing because the Ramanujan local sign generically neutralizes  $\mu$ . Thus the current target is a one-level signed Möbius-primitive-frequency dispersion theorem rather than another generic Kloosterman black box.

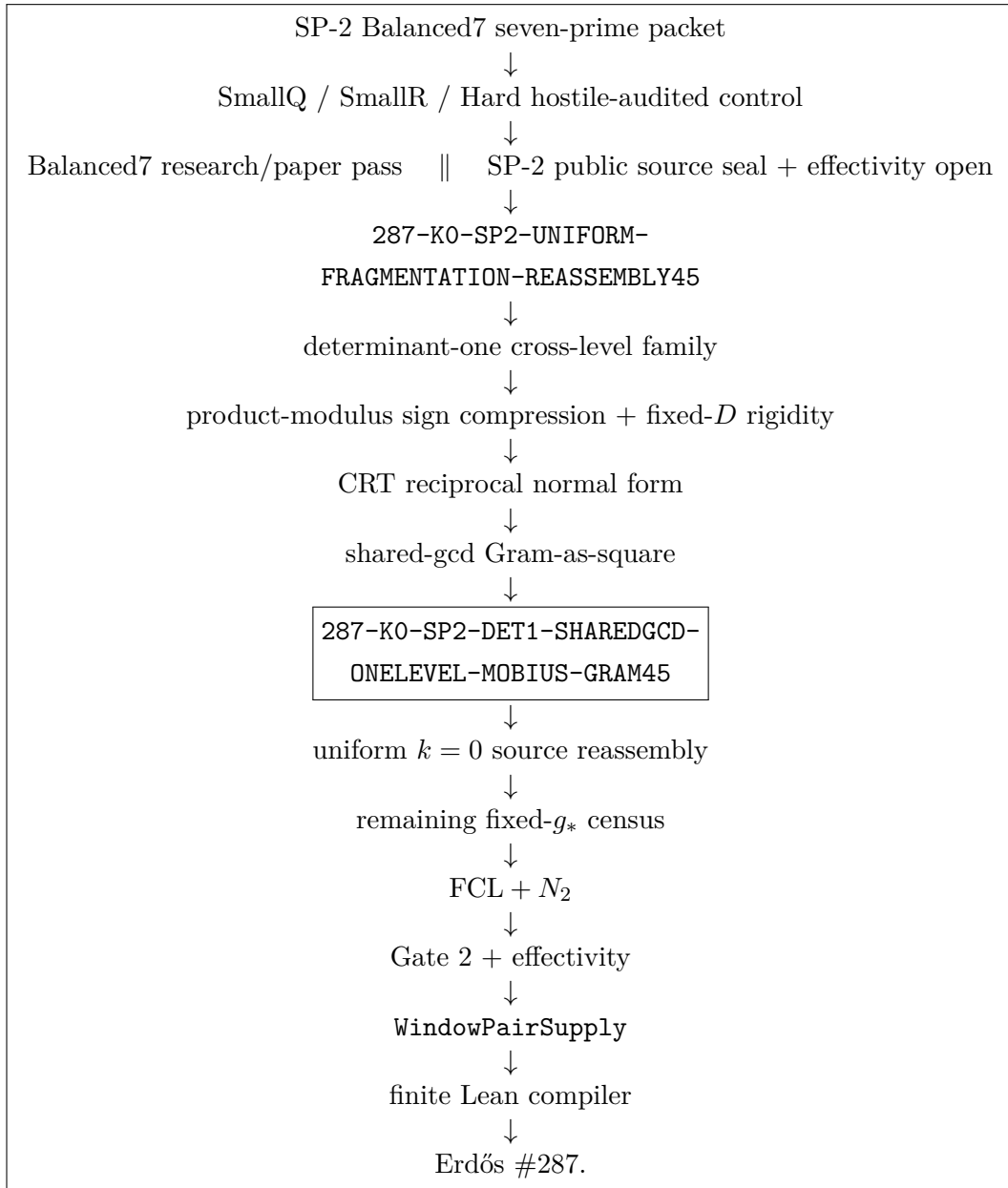
## 20 The current end-to-end dependency graph

The stable Gate spine is unchanged:



Provider assignment remains census-driven; the present #287 main line does not assume Gate 1A or Gate 1B globally.

The current #287 overlay is



Every arrow remains a firewall: a local research pass is not silently promoted to a source-exhaustive, effective or formal theorem.

## 21 Formal verification and reproducibility

### 21.1 Public manuscript repository

The public aggregate formal build presently imports the V15–V19 banks. Their supplied reports record successful builds of 8118, 8123, 8131, 8135 and 8140 jobs, respectively, with the stated zero-error and no-user-axiom firewalls. These banks cover, among other things,  $\Lambda = \mu * \log$  source algebra, factorial Euler polarization, labelled 3221 regrouping, finite diagonal/off-diagonal/no-wrap compilers, pre-Poisson divisibility and high-conductor character-Gram infrastructure. They do not prove the external analytic estimates used by the hostile research audits.

## 21.2 Later supplied safe bank

A later supplied append-only safe-bank report records a post-Balanced7 status/axiom audit and modules

- `SevenBoxPrimeWeights`; `PrimeTupleMultiplicity`;
- `ConductorSplitLargeSieve`; `LowConductorSiegelWalfisz`;
- `SmallQ34LSCompiler`; `SmallPrimePrefix`;
- `PostRepairOwnerCompiler`; `UniformFragmentationCompiler`.

The reported build has 8213 jobs and zero errors. Those modules are not present in the current public manuscript repository at this audit cutoff. R7 therefore labels them

SUPPLIED LEAN SAFE BANK / REPOSITORY SYNC PENDING,

not PUBLIC-MAIN SYNCHRONIZED.

## 21.3 Determinant-one formalization generations

Already supplied/formally banked finite material includes primitive Ramanujan algebra and reassembly, the short-lift finite profile, harmonic twists, shared- $g_0$   $u$ -parameter routing, averaged  $b$ -pair gcd identities, reduced-denominator algebra and exact near-frequency counting. Their analytic inputs remain uninhabited.

The newest product-modulus sign compression, fixed- $D$  frequency rigidity, CRT  $n$ -reciprocal normal form and shared-gcd Gram-as-square are

latest research-exact layer / formal banking pending

unless and until a successful corresponding formal replay is supplied and synchronized. R7 does not infer Lean verification from research nomenclature.

## 21.4 External analytic firewall

The Balanced7 SmallQ large-sieve, SmallR estimate, one-dimensional upper sieve, Shiu divisor average and short-lift zero-free-region inputs are external analytic results or research-audited applications. A Lean compiler may prove implications from named inputs; it does not thereby formalize the external theorem. Likewise the hostile-audited Balanced7 research/paper pass is not labelled a Lean analytic theorem.

The first exact one-level Möbius Gram is presently an open research theorem; no interface is treated as inhabited merely because the underlying exact algebra has been reduced.

## 22 Conclusion

The strongest unconditional result remains the public machine-checked finite theorem: no exact counterexample exists with maximum denominator at most  $4 \times 10^9$ . The exact global large- $M$  interface remains `WindowPairSupply`; no eventual effective supply theorem is proved.

The Balanced7 fixed-certificate seven-prime packet has now passed the completed hostile analytic audit at the research/paper level. The direct source has  $H_*(P) = -20$ , the SmallQ conductor-split  $3 + 4$  large-sieve route passes, the general-modulus SmallR owner is controlled, the physical hard range has the fixed margin  $\delta = 1/21$ , and the full- $q$  principal/defect ownership reassembles without double spending. This verdict remains subject to the exact public SP-2 source seal and is not an effective or Lean analytic closure. In particular, the classical low-conductor step does not provide a computable threshold.

The controlling main line has therefore moved to uniform  $k = 0$ . Two former global closure claims are withdrawn: large shared- $g_0$  cells and primitive near-frequency cells are not globally closed

merely from their local reciprocal-density gains once the full cross-level  $G^2/H$  multiplicity and all normalizations are restored.

The latest exact reorganization is more structural. Product-modulus compression replaces the two level signs by one  $\mu(n)$  sign; fixed- $D$  rigidity freezes the primitive frequency residue classes; CRT gives a squarefree unitary-divisor reciprocal kernel; and the shared-gcd projector converts the signed cross-level Gram exactly into

$$Q_H = \sum_d \frac{\lambda_H(d)}{d^2} \sum_a \rho(a) \left| \sum_m \frac{\mu(m)}{m} V_{dm}(a) \right|^2.$$

The apparent two-level signed family is therefore a one-level Möbius operator. A coefficient-blind estimate presently closes only a strict subpolytope and is provisional. Complete primitive-frequency Ramanujan reassembly is nonclosing because it neutralizes the remaining Möbius sign.

Accordingly the historical R7 first exact main-line analytic residual was

$$\boxed{287\text{-}K0\text{-}SP2\text{-}DET1\text{-}SHAREDGCD\text{-}ONELEVEL\text{-}MOBIUS\text{-}GRAM45.}$$

Its task is to exploit the one-level Möbius sign while preserving primitive frequency, the reciprocal  $b$  source, the friable  $a$  average and the exact shared-gcd projector.

Uniform  $k = 0$  remains open. FCL has not yet been reached;  $N_2$  and Gate 2 are not activated; effectivity remains open; eventual WindowPairSupply remains open.

$$\boxed{\text{Erdős Problem \#287 remains open.}}$$

## Part II

# Current Frontier and Provenance Ledger

## 23 Purpose, coverage baseline and status firewall

Part II restores the *technical depth* of the earlier public dossier without restoring its chronological patch-stack format. Its coverage floor is the V15 public-review snapshot committed as

7fc3797105519c4e6a1669f174ce7394336bdc18,

which published the former 50-page-style Gate-1A/Gate-1B/Ford programme together with the then-current balanced-seven update. That snapshot is used here only as a coverage baseline: when later work changes the controlling route, the later status wins.

Three time/status layers are kept separate throughout:

1. **public/formal bank:** theorem or compiler actually present in the public Lean project;
2. **latest supplied research ledger:** a mathematically audited frontier supplied after the V15 snapshot but not silently promoted to a public Lean or published theorem;
3. **open/source interfaces:** deliberately unproved analytic or source-identification obligations.

This is important for Gate 1B. The public formal socket still records the one-completion coefficient

$$\beta_{D,P}(d, p) = \mu_D(d) \Lambda_P(p)$$

and the exact 1/104 distribution-margin budget. That socket is neither false nor erased. However, the latest supplied analytic work has moved beyond it to a Motohashi/rank-one frontier. The formal socket is therefore a *banked abstraction/alternative provider interface*, while the rank-one chain below is the current research frontier.

Detailed dead-route archaeology remains outside the main narrative. What is restored here is every current component needed to reconstruct the proof programme: Gate 1A source class, Gate 1B current frontier and downstream firewall, the Ford-generated packet census, balanced- $R_9$ /fixed-certificate leakage, Gate 2, and the relation of the current factorial- $\Omega_7$  source to those providers.

## 24 Gate 1A: current canonical-provider ledger

Gate 1A is a conditional theorem for one canonical common-weight physical operator. It is not a universal theorem for arbitrary row-dependent source weights and it is not assumed globally necessary for Erdős #287.

For a row

$$e = (r, k, m), \quad m' = m + kr,$$

the affine forms satisfy

$$m' A_e(t) - m B_e(t) = 2k,$$

and the canonical physical operator is

$$C_e(b, d) = \sum_t W_D(t) \left( \sum_{p \preceq L} b_p \rho_p(A_e(t)) \right) \left( \sum_{q \preceq L} d_q \rho_q(B_e(t)) \right).$$

The conditional target remains

$$\sum_{e \in \mathcal{E}^\#} |C_e(b, d)|^2 \ll \frac{ML^4}{H} X_1^{o(1)} \sum_{p \preceq L} |b_p|^2 \sum_{q \preceq L} |d_q|^2,$$

with the same normalization conventions as in Part I.



## 24.1 What is banked and what is still external

| Gate-1A component                                              | Status                    | Current interpretation                                                                                  |
|----------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------|
| Determinant/row algebra and all- $m$ row family                | BANKED FINITE/OPERATOR    | The theorem is not restricted to clean- $P_3$ rows.                                                     |
| Canonical common physical weight $W_D$                         | SOURCE FIREWALL BANKED    | Later row dependence must be deterministically derived from this one datum.                             |
| Scale ledger, row-energy conservation and normalization bridge | BANKED                    | Includes the $H^2$ conversion and the signed vertex margins.                                            |
| Finite BPP participation-to-energy compiler                    | BANKED                    | Participation implies a moving-family energy saving once the physical participation input is supplied.  |
| Projective ratio-class/axis algebra                            | BANKED FINITE ALGEBRA     | The state diagonal is routed inside the zero-curvature/projective branch rather than discarded.         |
| Outer-square-root algebra and exponent budget                  | BANKED                    | The formal compiler checks the square-root conversion but does not manufacture the physical energy pin. |
| Poisson–Bruhat transport/source ceiling                        | SOURCE PIN                | Must preserve the literal physical packet and absolute comparator.                                      |
| Fixed-state exclusion and prime participation                  | EXTERNAL PIN              | The finite compiler is present; the physical smooth-envelope/short-interval input is external.          |
| Outer-root physical energy identification                      | SOURCE OPEN               | Identifies the abstract BPP energy with the actual Gate energy.                                         |
| Projective/diagonal physical source routing                    | SOURCE OPEN               | Finite algebra exists; the physical source ceiling must be supplied at target scale.                    |
| Literal recombination estimate                                 | SOURCE OPEN               | Controls discarded/excision/recombination errors without spending the same saving twice.                |
| Arbitrary edge-dependent $D_2$ family                          | OUTSIDE CANONICAL THEOREM | Requires an upstream Gate-0 adapter or finite-template/nuclear reduction.                               |

The current source-glue issue can be summarized as a synchronized two-view/fibre problem: the same physical packet has to satisfy the common-weight representation, the BPP participation representation and the final source-routing representation simultaneously. We refer to this compactly as the *SB- $\nu$ /source-glue interface*. It is a source-identification problem, not an exponent deficit in the already-banked finite operator algebra.

## 24.2 When Gate 1A is required for #287

The correct logical statement is

GATE1A\_REQUIRED is decided by the literal Ford-generated packet census.

An actual generated packet may use Gate 1A only after a scale/source dictionary supplies:

- the relation between the Gate ambient scale  $X_1$  and the #287 scale  $X$ ;

- the identification of the physical bilinear variables with the Gate row variables;
- the induced  $R, L, H, K, U, D$  ranges;
- preservation of the one common physical weight, or a proved finite-template/nuclear reduction with acceptable total cost;
- survival of the binding fixed-power margin after truncation, multiplicity and scale conversion.

No global Gate-1A closure is assumed if the generated census never routes a needed packet to this source class.

## 25 Gate 1B: banked socket and current Bézout-dual frontier

Gate 1B remains the complementary signed/parity-breaking provider. Its banked one-completion socket

$$\beta_{D,P}(d,p) = \mu_D(d)\Lambda_P(p)$$

and the exact  $1/104$  distribution-level capacity ledger remain valid, but they are not the current first research residual.

The historical progression through the Motohashi/fivefold source, rank-one endpoint and weighted shifted top-band repair is retained as provenance. The current public research index instead records the strict reduction

$$\text{C4SHIFT-OFFDIAG-CENTERED-AP5/8-GRAM45} \implies \boxed{\text{C4SHIFT-BEZOUT-DUAL-ROELL45}}.$$

On  $(d, \ell) = 1$ ,  $r = dr_0$ , the Bézout line

$$bZ_2 - aZ_1 = \ell r_0$$

separates as

$$Z_1 = \ell r_0 \kappa_{a,b} + bt, \quad Z_2 = \ell r_0 \bar{b}_a + at,$$

with physical residue

$$t \equiv -2\overline{usdab} \pmod{\ell}$$

independent of  $r_0$ . The centred cyclic dual kernel is

$$\widehat{K}_{\text{cent}}(\nu) = \ell^{-1} e_\ell(-\nu\tau) \mathfrak{q}(\nu), \quad \mathfrak{q}(\nu) = 1 - \ell^{-1} \sum_k \mathcal{M}_1(k + \nu \overline{dab}) \overline{\mathcal{M}_2(k)}.$$

The zero mode is a local owner and feeds `C4SHIFT-BEZOUT-ZEROMODE-LOCALMATCH45`; it is not estimated away. For  $\nu \neq 0$ , after opening the gamma source,

$$bx_2y_2 - ax_1y_1 = \ell r_0, \quad e_\ell(\nu t) = e_\ell(\nu \bar{b} x_1 y_1).$$

The provisional local census bound

$$\mathcal{N}_D \ll Y^{3/4} \sqrt{\min(D, T/D)} L^C, \quad T = Y^{3/2} L^{O(1)},$$

is `HOSTILE-AUDIT PENDING`. It suggests boundary-band closure and reduces the worst residual to  $Y^{3/8} = X^{1/24}$ , but the central  $D$  range remains and the claimed fixed-fibre uniqueness must be checked allowing  $t$ -differences by multiples of  $\ell$ .

Accordingly

$$\boxed{\text{C4SHIFT-BEZOUT-DUAL-ROELL45}}$$

is the first exact Gate-1B analytic residual at research level. The missing mechanism is source-specific cancellation in the coupled nonzero  $(\nu, r_0, \ell)$  Bézout/gamma family. The parallel zero-mode/top-band local comparison is independently source-open.

### 25.1 Downstream Gate-1B firewall

Only after the nonzero Bézout child and the local zero-mode/top-band comparison both close may the compiler promote the centred-AP parent, one-minor pushed energy and Q-Fourier pushforward. These then feed the remaining beta/determinant/top-band descendants, Pure5 defect strata, NearPrim, the  $r > 1$  recursion, QK56 exhaustiveness and final source reassembly. No earlier node is allowed to skip these descendants.

## 26 Ford–Maynard generated packets: census before provider

The Ford–Maynard universal Type-II hypothesis is a clean published sufficient wrapper, but Proposition 7.22 does not invoke arbitrary coefficient pairs out of thin air. After a finite decomposition it generates the grouped equation-(7.23) family

$$\sum_{n=n'n''\asymp x} Y_1(n')Y_2(n'')w_{n'n''}.$$

A proof-specific replacement is therefore legitimate only after two source obligations:

1. a *literal generated-packet census* listing every coefficient family produced by the proof;
2. a replacement certificate  $C_{\text{FM}}$  proving that every Type-II invocation is covered by one proved provider after all Mellin/Perron, support and norm transformations.

The programme labels

SOURCE-ADAPTER45,  
FM-TO-GATE-COORDINATE-CENSUS45

refer to this source dictionary. Neither can be replaced by the slogan “the packet looks Type-II”.

### 26.1 Current provider-facing census

| Generated family               | Literal status                   | Candidate provider                     | Current blocker                                                          |
|--------------------------------|----------------------------------|----------------------------------------|--------------------------------------------------------------------------|
| Möbius singleton               | source-specific generated packet | Gate 1B / signed source                | literal dictionary and stability under generated outer factors           |
| Model singleton                | source-specific generated packet | QK56/direct affine completion          | native packet dictionary; short Möbius carrier may sit in the complement |
| Balanced-seven V-cell          | explicit fixed-degree source     | factorial- $\Omega_7$ endpoint theorem | signed endpoint + physical comparison identification                     |
| Balanced $R_9$ leakage         | Ford-generated leakage candidate | fixed-certificate/FCL lane             | literal $G(d; n)$ specialization and source reassembly                   |
| Common-weight/edge packets     | census-dependent                 | Gate 1A only after exact match         | scale/source adapter and common-weight preservation                      |
| Pure-five/high-order packets   | Gate-1B source family            | Motohashi/rank-one chain               | endpoint off-diagonal, comparison, high- $k$ , downstream strata         |
| Remaining fixed- $g_*$ packets | not exhaustively censused        | TBD from literal packet shape          | finite census and provider assignment                                    |

This table encodes the main logical correction:

provider assignment follows the generated source; it is not imposed globally.

In particular, Gate 1A is neither deleted nor assumed universally necessary.

## 27 Balanced $R_9$ , fixed-certificate leakage and the Gate-2 landing

The balanced order-nine identity

$$\sum_{j=0}^4 (-1)^j \binom{9}{j} = 70$$

is a genuine finite/binomial fact. The formal bank also records the complementary alternating identity. What is *not* automatically proved is that Ford's actual arithmetic weight  $G(d; n)$  selects exactly this divisor-order specialization on the physical balanced- $R_9$  cell.

Therefore the correct status is:

balanced  $R_9$  = Ford-generated leakage candidate,

with

$H(n) = 70$  only after the literal  $G(d; n)$  specialization is proved.

The old  $R_9$  death certificate and the abandoned unbalanced-1/5 redesign are not revived.

The fixed-certificate architecture uses one explicit positive Ford certificate  $g_*$  and its arithmetic weight

$$H_*(n) = \sum_{d|n} G_*(d; n).$$

Its transference identity separates:

- total correlation;
- leakage outside the prime/main-composite regions;
- the exceptional  $N_2$  collar;
- the positive comparison margin.

Balanced  $R_9$  is one generated leakage family that can feed this architecture once the physical source dictionary is proved; it is not the whole fixed-certificate leakage problem.

### 27.1 Gate 2 consumes, rather than invents, the packet bounds

Gate 2 is the Ford–Maynard lower-bound/endgame compiler. It fixes one compatible  $\varepsilon_*$  and the shrunk parameter triple

$$P_{\varepsilon_*} = \left( \frac{1}{2} - \varepsilon_*, \varepsilon_*, \frac{1}{6} - 2\varepsilon_* \right).$$

Its source-specific replay requires:

1. Gate-0 comparison regularity, growth and exact Type-I input;
2. the literal  $G(d; n)/H(n)$  dictionary and the Ford-generated packet census;
3. the source-specific aggregate  $N_2$  estimate;
4. one compatible fixed  $\varepsilon_*$ ;
5. either the universal Type-II wrapper, or all generated-(7.23) packet estimates together with SOURCE-ADAPTER45 and  $C_{\text{FM}}$ .

Thus the correct closure order is

packet census  $\rightarrow$  Gate providers  $\rightarrow$  generated Type-II reassembly  $\rightarrow$  Gate 2 lower bound,  
not Gate 2 followed by a later packet census.

## 28 Current #287 frontier after Balanced7

The fixed-degree-seven Balanced7 packet is now an audited input to the next source layer, not the first main-line analytic residual. Its status is:

HOSTILE-AUDITED RESEARCH/PAPER PASS  
SUBJECT TO EXACT SP-2 SOURCE SEAL

The effective version is separately open.

The next parent is

287-K0-SP2-UNIFORM-FRAGMENTATION-REASSEMBLY45.

Within its determinant-one cross-level sector the latest exact reductions are:

1. short-lift Euler collapse after a uniformity repair;
2. product-modulus compression  $\mu(g_1)\mu(g_2) = \mu(r_1r_2)$ ;
3. fixed- $D$  frequency rigidity and  $t_1/g_1 - t_2/g_2 = D/(g_0n)$ ;
4. CRT reciprocal normal form modulo  $n$ ;
5. two-state prime assignment at fixed  $n$ , with an explicit firewall against falsely promoting it to multiplicativity in  $n$ ;
6. exact shared-gcd Gram-as-square;
7. a provisional coefficient-blind strict-subpolytope energy bound;
8. a negative Ramanujan-completion result showing why the Möbius sign must be used before full primitive-frequency completion.

The controlling one-level object is

$$\mathcal{G}_{H,d} = \sum_{a \succ A} \rho(a) \left| \sum_{\substack{m \succ G/d \\ (m,d)=1}} \frac{\mu(m)}{m} \sum_{t \bmod dm}^* \tilde{G}_{dm,s,\tau}(t) e_{dm}(ta) \right|^2,$$

and the current first exact main-line target is

$$\left| \sum_d \frac{\lambda_H(d)}{d^2} \mathcal{G}_{H,d} \right| \ll AB^2 (\log X)^{-6-2\eta-C_{\text{route}}}.$$

Programme label:

287-K0-SP2-DET1-SHAREDGCD-ONELEVEL-MOBIUS-GRAM45.

This target remains open.

| Node | Current status | Publication interpretation |
|------|----------------|----------------------------|
|------|----------------|----------------------------|

## 29 Master current-status ledger

| Node                                                                 | Current status                                                | Publication interpretation                                                                                                                              |
|----------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Finite exclusion $M \leq 4 \times 10^9$<br>WindowPairSupply compiler | PUBLIC LEAN-CHECKED<br>PUBLIC LEAN-CHECKED<br>/ CONDITIONAL   | Strongest unconditional theorem.<br>Global closing compiler; eventual<br>supply open.                                                                   |
| Gate 0                                                               | MIXED PASS / SOURCE<br>PINS                                   | Physical comparison/Type-I<br>package.                                                                                                                  |
| Gate 1A                                                              | CONDITIONAL                                                   | Use is packet-census dependent.                                                                                                                         |
| Gate 1B                                                              | CANONICAL PROVIDER<br>OPEN                                    | First exact research residual<br>C4SHIFT-BEZOUT-DUAL-ROELL45;<br>feeding $D$ -census hostile-audit<br>pending and zero-mode local<br>match source-open. |
| Balanced7 full- $q$ analytic packet                                  | HOSTILE-AUDITED<br>RESEARCH/PAPER PASS                        | Conditional on exact public SP-2<br>source seal; not Lean analytic.                                                                                     |
| Balanced7 source seal                                                | SOURCE-OPEN IN PUBLIC<br>MANUSCRIPT REPO                      | Later supplied safe bank not<br>synchronized here.                                                                                                      |
| Balanced7 effective                                                  | OPEN                                                          | Effective<br>low-conductor/exceptional PNT<br>dictionary missing.                                                                                       |
| 287-K0-SP2-UNIFORM-<br>FRAGMENTATION-REASSEMBLY45                    | OPEN / HISTORICAL R7<br>PARENT / SUPERSEDED<br>AS CONTROLLING | Must cover actual $k$ -range and all<br>fragmentation/source states.                                                                                    |
| DET1-LARGESHAREDGO-CELLS45 global<br>closure                         | RETRACTED                                                     | Local fixed-pair gains survive; full<br>cross-level multiplicity was<br>missing.                                                                        |
| DET1-PRIMITIVE-NEARFREQ45 global<br>closure                          | RETRACTED                                                     | Near-density gain survives locally<br>but does not close the global<br>family.                                                                          |
| DET1-SHORTLIFT-EULER-COLLAPSE45                                      | PASS AFTER<br>UNIFORMITY REPAIR                               | Analytic zero-free input not Lean<br>formalized.                                                                                                        |
| DET1-LEVELPAIR-PRODUCTMOD-SIGN45                                     | RESEARCH-EXACT PASS<br>/ BANK PENDING                         | Two level signs compress to one<br>$\mu(n)$ .                                                                                                           |
| DET1-FIXEDD-FREQUENCY-RIGIDITY45                                     | RESEARCH-EXACT PASS<br>/ BANK PENDING                         | Fixes primitive residues; friable<br>kernel depends on $D/(g_0n)$ .                                                                                     |
| DET1-LEVELPAIR-N-<br>RECIPROCAL-NORMALFORM45                         | RESEARCH-EXACT PASS<br>/ BANK PENDING                         | CRT reciprocal source modulo $n$ .                                                                                                                      |
| DET1-LEVELPAIR-PRIMEASSIGNMENT45                                     | PASS AT FIXED $n$                                             | Does not imply multiplicativity in<br>$n$ .                                                                                                             |
| DET1-LEVELPAIR-SINGLEMOBIUS-N45                                      | OPEN                                                          | Ordinary Euler-product route<br>unavailable.                                                                                                            |
| DET1-SHAREDGCD-GRAM-SQUARE45                                         | RESEARCH-EXACT PASS<br>/ BANK PENDING                         | Compresses cross-level Gram to<br>one-level Möbius operator.                                                                                            |
| DET1-SHAREDGCD-ONELEVEL-ENERGY45                                     | PROVISIONAL<br>STRICT-SUBPOLYTOPE<br>CLOSURE                  | Hostile promotion audit pending.                                                                                                                        |
| ONELEVEL-RAMANUJAN-ZEROFREE45                                        | NONCLOSING /<br>REPRESENTATION LOOP                           | Full Ramanujan completion<br>neutralizes Möbius sign.                                                                                                   |
| 287-K0-SP2-DET1-SHAREDGCD-<br>ONELEVEL-MOBIUS-GRAM45                 | FIRST EXACT MAIN-LINE<br>ANALYTIC RESIDUAL /<br>OPEN          | Exploit one-level signed Möbius<br>operator before full primitive<br>completion.                                                                        |
| FCL                                                                  | NOT REACHED / OPEN                                            | Downstream of uniform $k = 0$ and<br>remaining fixed-certificate census.                                                                                |
| $N_2$ / Gate 2                                                       | NOT ACTIVATED                                                 | Downstream.                                                                                                                                             |
| 287-EFFECTIVE-LOWCOND-EXCEPTIONAL-<br>PNT45                          | OPEN                                                          | Effective threshold lane.                                                                                                                               |
| Eventual WindowPairSupply                                            | OPEN                                                          | Required before finite compiler<br>fires globally.                                                                                                      |

| Node               | Current status | Publication interpretation    |
|--------------------|----------------|-------------------------------|
| Erdős Problem #287 | OPEN           | No unconditional proof claim. |

### 30 Superseded, false and provenance-only routes

The project preserves failed and superseded routes because they explain the source firewalls, but they do not control R7:

- squarefree polarization plus the  $X^{6/7+o(1)}$  repeated-prime router: valid historical encoding, superseded by factorial Euler polarization;
- complete multiplicativity as a required repair: retired/not needed;
- the direct Pascadi smooth-number splice at  $(3/5, 1/7)$ : nonclosing under the audited written parameters;
- V14 Vaughan/prime-modulus two-outer route: valid but nonminimal source audit;
- direct singleton = existing common-conductor Gate-1B covariance: literal source dictionary false;
- direct Gate-1A determinant-one adapter: literal determinant mismatch;
- equal-seven P6.3 three-block route and separated-large-sieve closure: nonclosing;
- DI/Kuznetsov and later HHH-first 3221 lanes: preserved as formal/source infrastructure but superseded after the direct full- $q$  SP-2 audit;
- unique Ford-(7.23) Balanced7 packet interpretation: retracted; the literal source is fixed-certificate SP-2;
- SmallQ Gate-0 Type-I: wrong provider for  $q \mid 2P + s$ , not a false theorem in its proper source class;
- large-shared- $g_0$  and primitive-near-frequency *global closure*: retracted after restoring the  $G^2/H$  cross-level multiplicity and full normalizations; their local fixed-pair cancellation remains valid;
- PRIMITIVE-SMALLGCD-FAR-HARDDEN-GRAM45: proposed but never promoted as the controlling residual;
- ordinary Euler-product cancellation from the fixed- $n$  two-state assignment: unavailable because the local factor depends on the complementary coprime component;
- complete primitive- $t$  Ramanujan zero-free route: nonclosing because its local sign generically neutralizes the Möbius sign.

### 31 Formal theorem and provenance map

#### Public Lean bank

The public manuscript repository contains the exact problem predicate, finite exclusion through  $4 \times 10^9$ , WindowPairSupply compiler, fixed-certificate source compilers, V15–V19 affine/factorial/3221 modules and their aggregate imports.

## Supplied later safe bank

A later supplied report records an 8213-job successful append-only build with seven-box weights, product multiplicity, conductor-split large-sieve compiler, low-conductor Siegel–Walfisz interface, SmallQ compiler, small-prime-prefix source, post-repair owner compiler, uniform-fragmentation compiler and post-Balanced7 status audit. Those modules are not currently present in this public manuscript repository. Their R7 status is SUPPLIED LEAN SAFE BANK / REPOSITORY SYNC PENDING.

## Already supplied determinant-one formal bank

The supplied formal work already banks finite primitive Ramanujan algebra/reassembly, short-lift finite profiles, harmonic twists, shared- $g_0$   $u$ -parameter routing, averaged  $b$ -pair gcd identities, reduced-denominator algebra and exact near-frequency counting. No deep analytic input is thereby inhabited.

## Latest research-exact layer, formal banking pending

The product-modulus Möbius-sign compression, fixed- $D$  frequency rigidity, CRT  $n$ -reciprocal normal form and shared-gcd Gram-as-square are currently research-exact mathematics. R7 does not call them Lean-verified until a corresponding successful formal bank is supplied and synchronized.

## External analytic boundary

The Balanced7 hostile audit uses external multiplicative large-sieve, Siegel–Walfisz, one-dimensional upper-sieve and Shiu inputs. The short-lift uniformity repair uses a zero-free-region analytic statement. The first one-level Möbius Gram remains open. Thus the formal compiler/source bank and the analytic research status remain deliberately separate.

## References

- [1] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monographies de L’Enseignement Mathématique, 1980, p. 33.
- [2] T. F. Bloom, *Erdős Problem #287*, Erdős Problems archive, problem provenance and status page, accessed August 2026.
- [3] K. Ford and J. Maynard, *On the theory of prime producing sieves*, arXiv:2407.14368, 2024.
- [4] S. Drappeau, A. Granville and X. Shao, *Smooth-supported multiplicative functions in arithmetic progressions beyond the  $x^{1/2}$ -barrier*, Mathematika 63 (2017), 895–918.
- [5] A. Pascadi, *On the exponents of distribution of primes and smooth numbers*, arXiv:2505.00653v2, 29 June 2025.
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## 31 August 2026 controlling final status

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|                                   |                                                      |
|-----------------------------------|------------------------------------------------------|
| Finite $M \leq 4 \times 10^9$     | KERNEL-PROVED / unchanged                            |
| C0 exact-product / Double Type II | ANALYTICALLY BANKED                                  |
| C0                                | ANALYTICALLY CLOSED / CONDITIONAL SOURCE PIN         |
| Formal source pin                 | SHAREDGCD-OMEGAH-NORMALISATION-SOURCEPIN45           |
| Full transverse branch            | STRICTLY REDUCED / OPEN                              |
| First transverse residual         | CRITICAL-BEZOUTNUMERATOR-SINGLECARRIER45             |
| Parallel $b$ -diagonal            | OPEN                                                 |
| Supplied latest formal build      | 8298 jobs / 0 errors; public dependency sync pending |
| Erdős #287                        | OPEN                                                 |

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The historical Gate 0–2 theory, R7 uniform- $k = 0$  branch, Ford/FCL architecture, and finite compiler are preserved above. No implication from the local C0/transverse milestones to a complete solution is asserted. Downstream Twin Prime work, if referenced, is a separate programme and is not proved by this manuscript.